# CHAPTER 96 — ELECTRICAL SYSTEM

## CONTENTS — MAINTENANCE PROCEDURES


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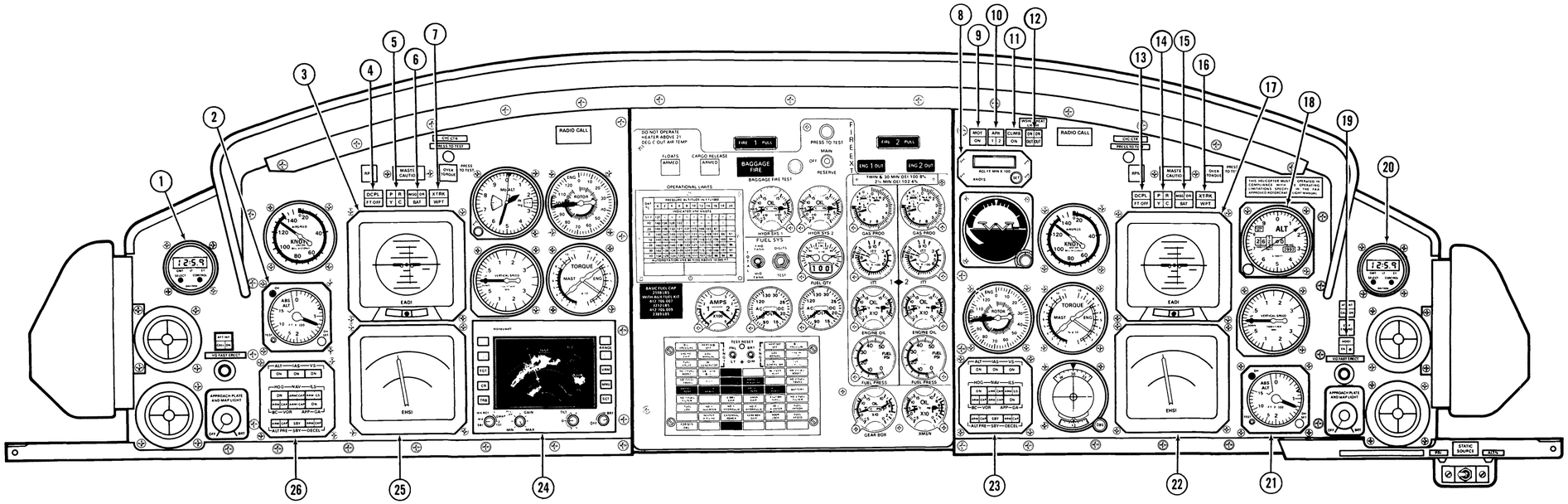
# ELECTRICAL SYSTEM

## ELECTRICAL SYSTEMS

### 96-1. ELECTRICAL SYSTEMS.

The supplemental electrical equipment installed in the search and rescue (SAR) configured model 412 helicopter consists of the dual digital automatic flight control system (DDAFCS), and the electronic flight instrument system (EFIS).

Components that are part of the electrical configuration on SAR helicopters are illustrated on figures 96-1 through 96-7. Refer to Chapter 98 for schematic and wiring diagrams. For code designation correlation, refer to Electrical Equipment List ( Table 96-1).



1. 5M5 Clock (copilot)
2. 3408DS1 Radio altimeter indicator (copilot)
3. 3440DS1 EADI (copilot)
4. 9DS32 Force trim OFF/flight director DCPL light (copilot)
5. 9DS31 Pitch, roll, yaw, collective out-of-detent light (copilot)
6. 3415DS1 MSG/DR/BAT light (copilot)
7. 3415DS3 XTRK/WPT light (copilot)
8. 9A6 Altitude preselect
9. 9DS36 MOT/ON light
10. 9DS38 APR/1/2 light
11. 9DS37 CLIMB/ON light
12. 10DS1 Windshield heater ON/OUT light
13. 9DS35 Force trim OFF/flight director DCPL light (pilot)
14. 9DS34 Pitch, roll, yaw, collective out-of-detent light (pilot)
15. 3415DS2 MSG/DR/BAT light (pilot)
16. 3415DS4 XTRK/WPT light (pilot)
17. 3440DS2 EADI (pilot)
18. 3410DS1 Encoding altimeter
19. 9DS39 HOIST/EN light
20. 4M5 Clock (pilot)
21. 3408DS2 Radio altimeter indicator (pilot)
22. 3440DS4 EHSI (pilot)
23. 9A4 Mode selector (pilot)
24. 3420DS1 Color indicator (radar indicator)
25. 3440DS3 EHSI (copilot)
26. 9A3 Mode selector (copilot)

412-SAR-96-1

**Figure 96-1. Instrument panel**

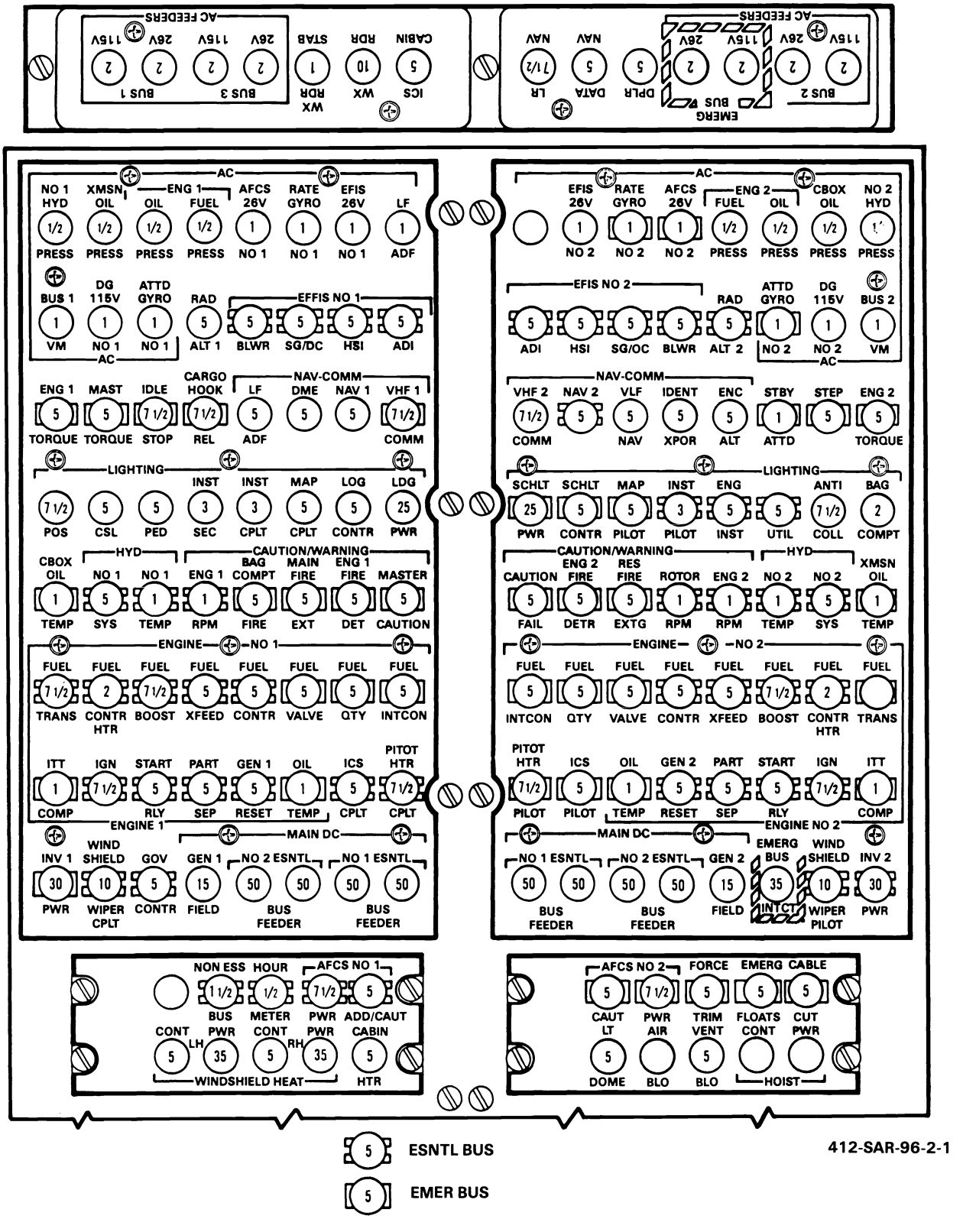
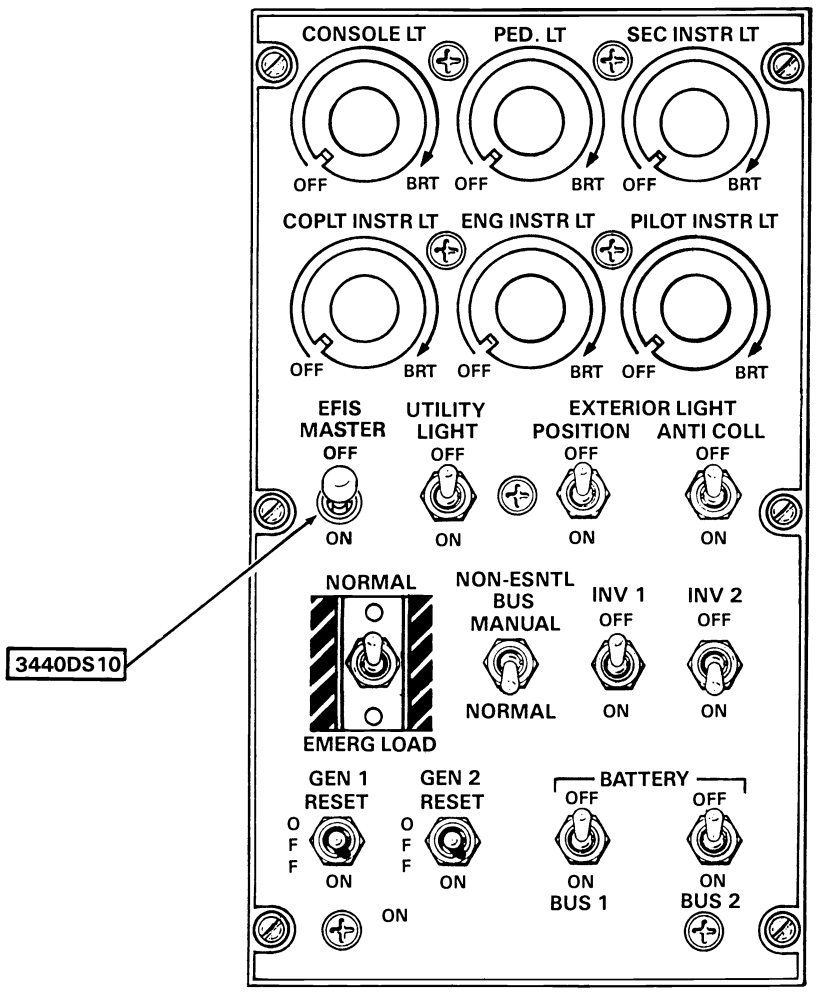


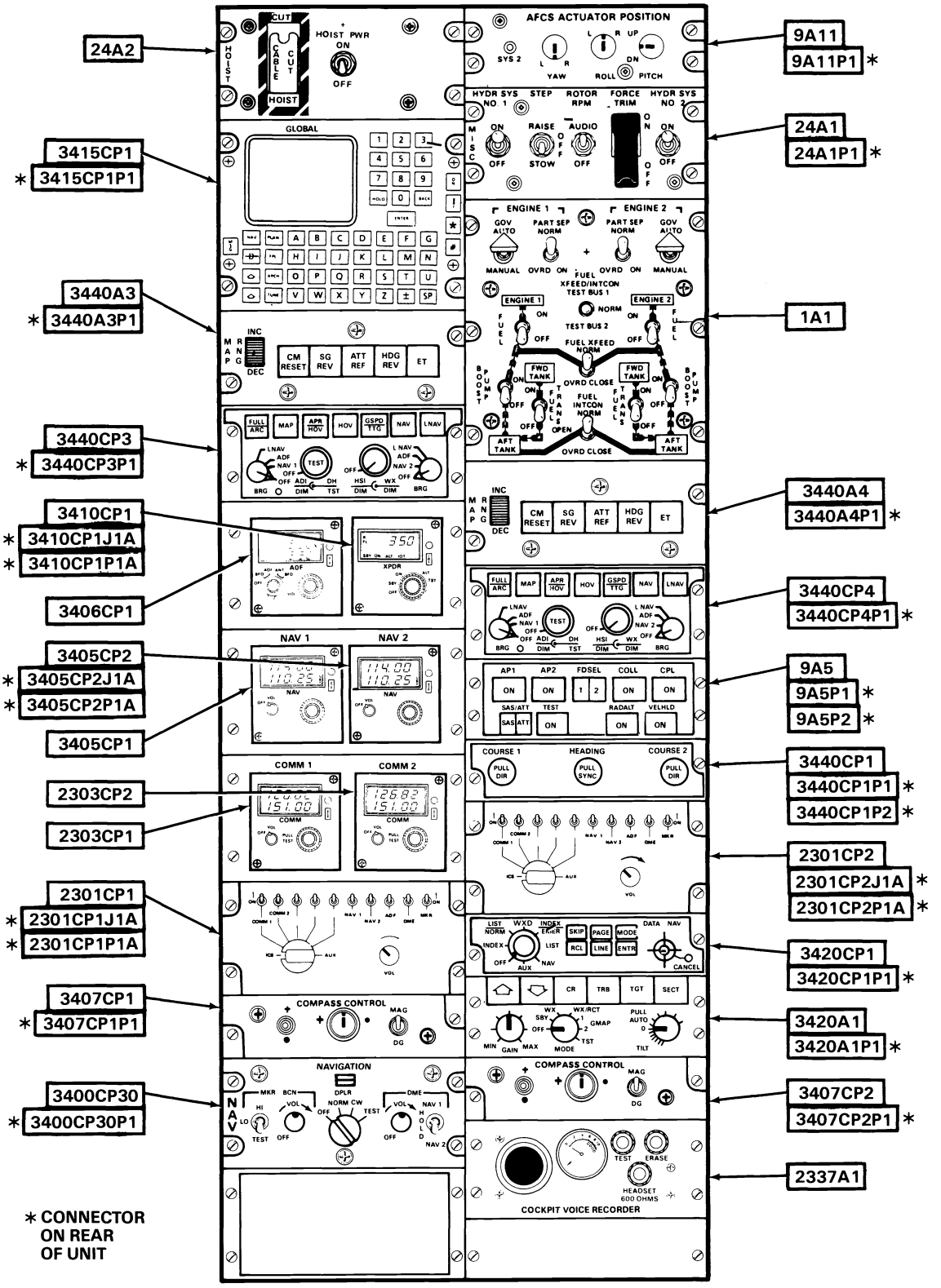

**Figure 96-2. Console (Sheet 1 of 2)**




OVERHEAD CONSOLE
PILOT SIDE

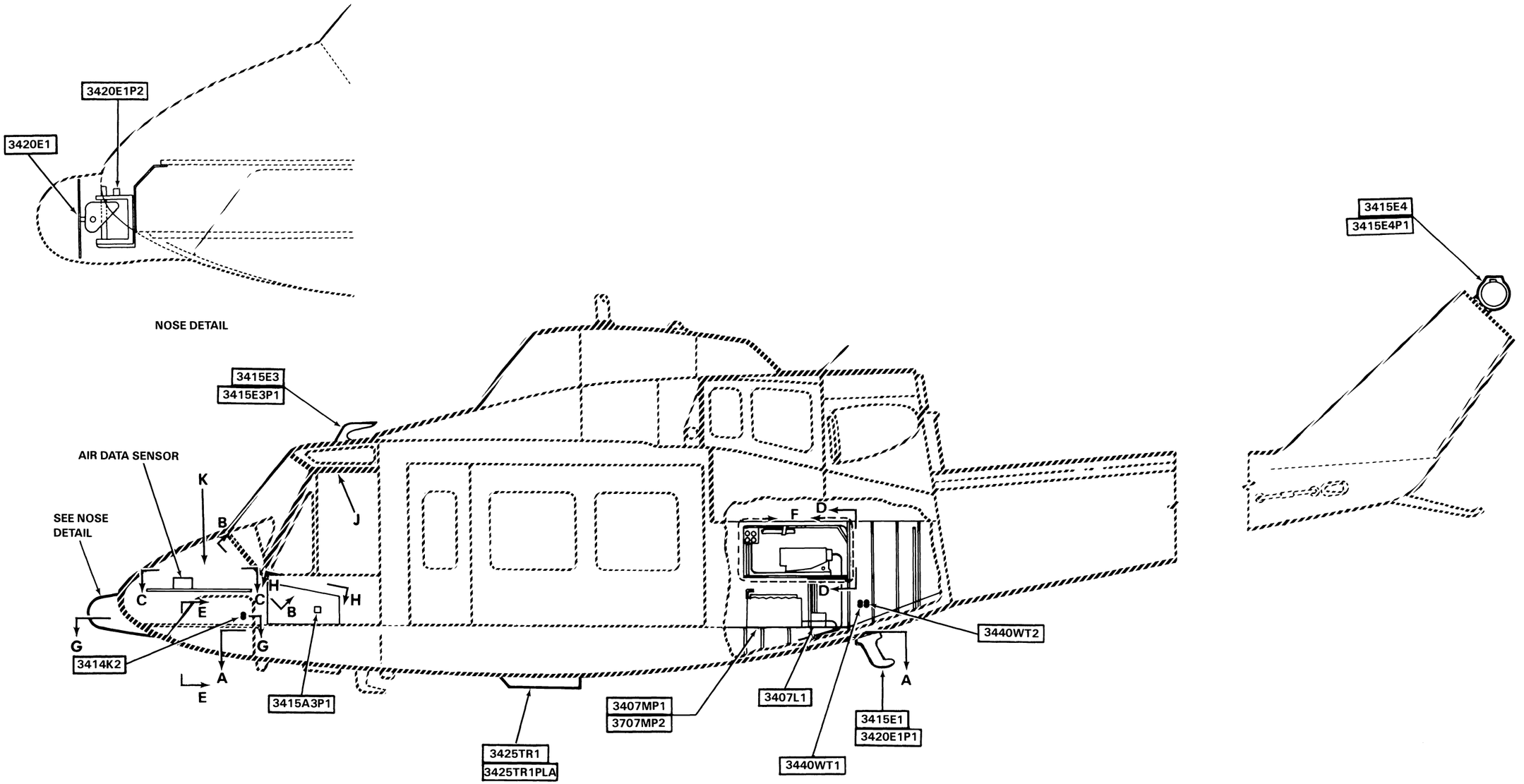
412-SAR-96-2-2

**Figure 96-2. Console (Sheet 2)**




412-SAR-96-3

**Figure 96-3. Pedestal**




412-SAR-96-4-1
412-575-007C

**Figure 96-4. EFIS related component location (Sheet 1 of 8)**

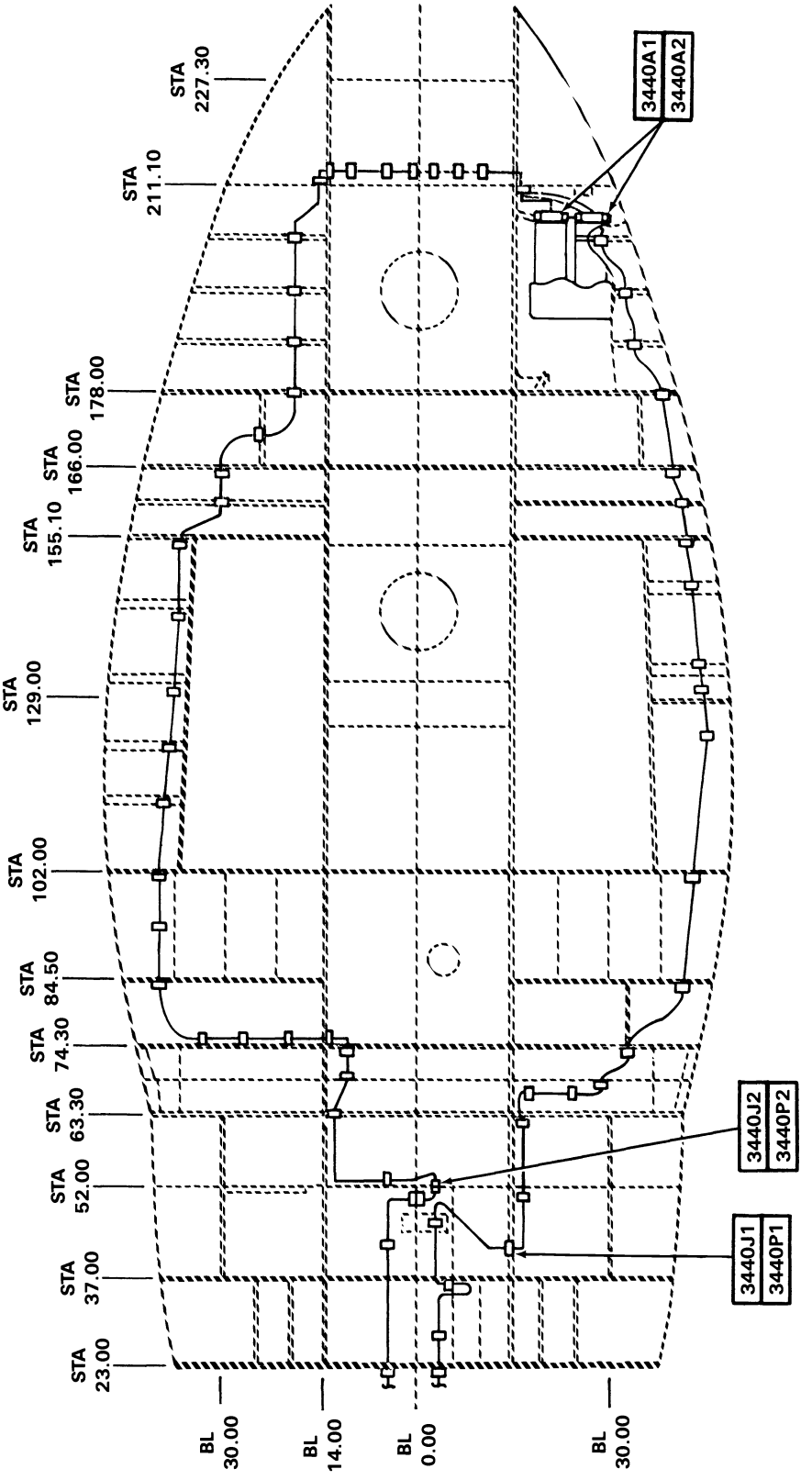



**Figure 96-4. EFIS related component location (Sheet 2)**

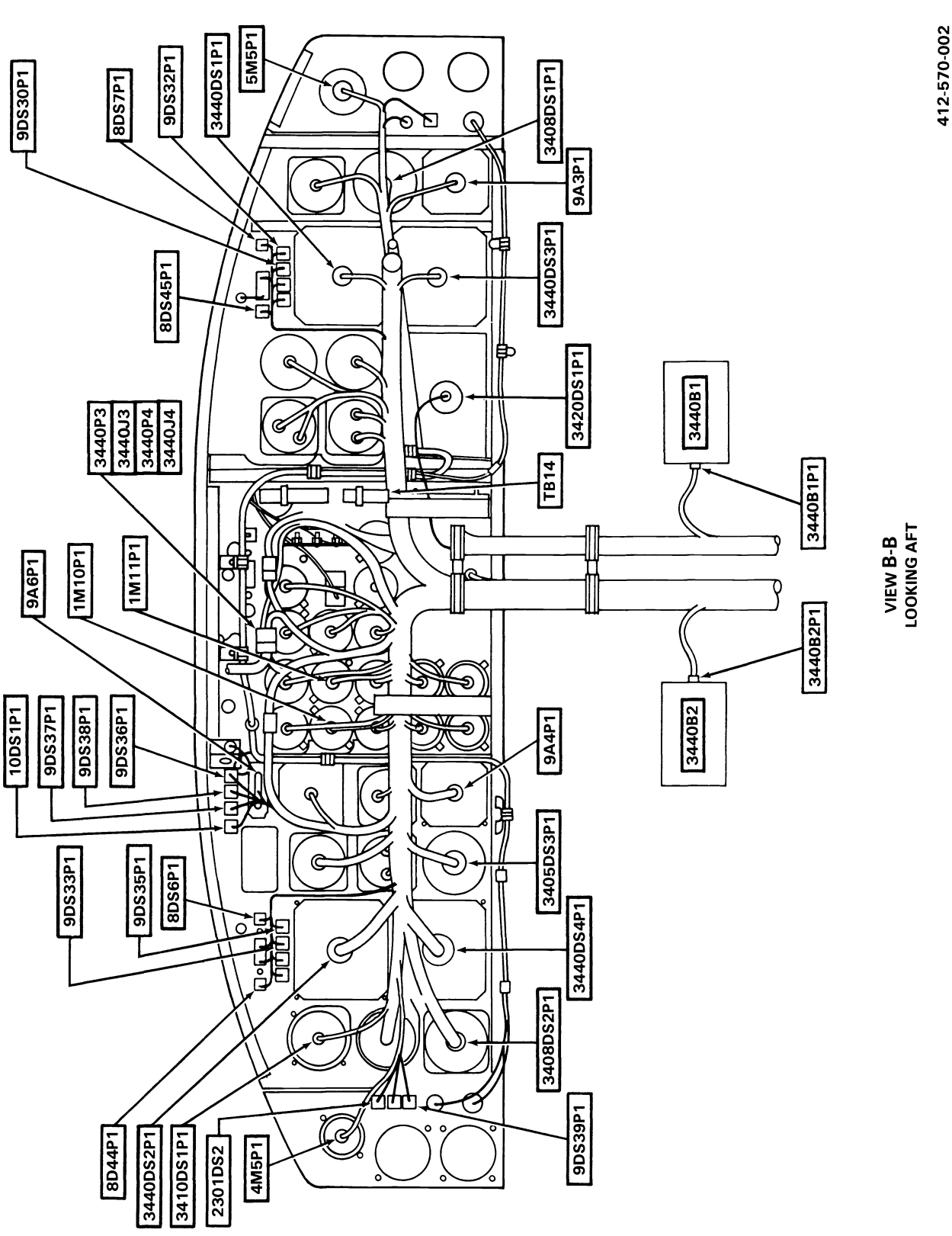
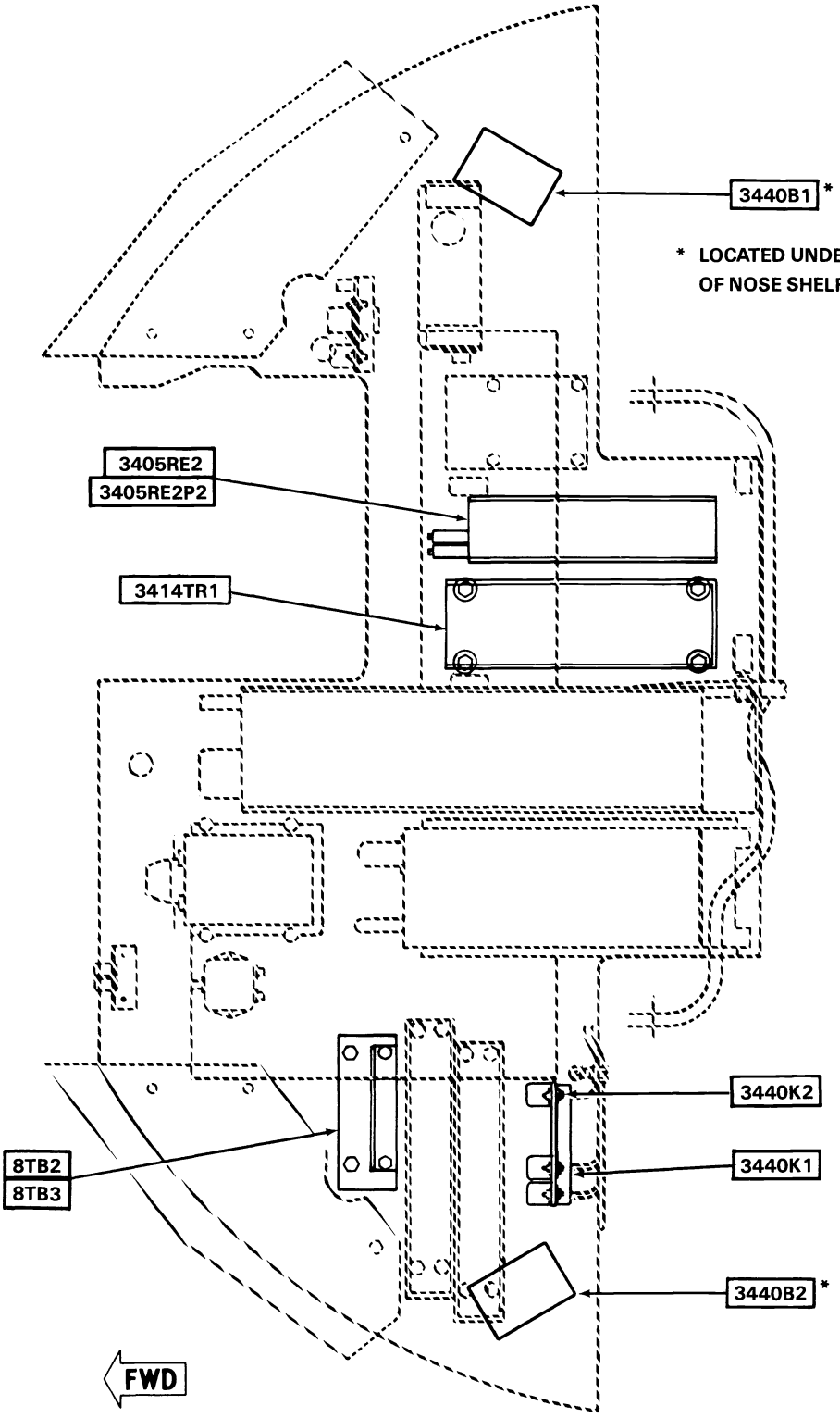



**Figure 96-4. EFIS related component location (Sheet 3)**




VIEW C-C
LOOKING DOWN AT UPPER NOSE SHELF

412-705-004
412-SAR-96-4-4

**Figure 96-4. EFIS related component location (Sheet 4)**

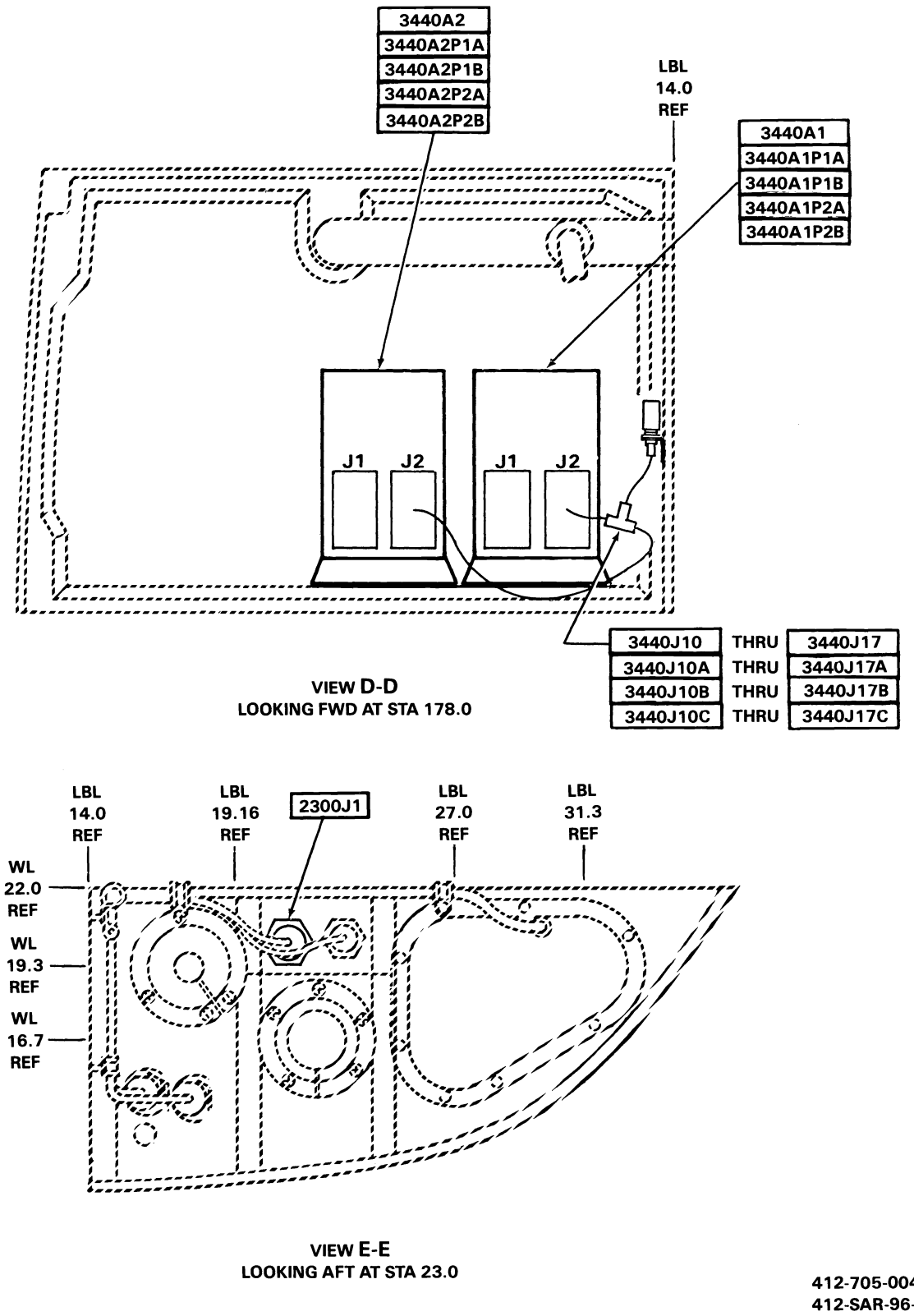
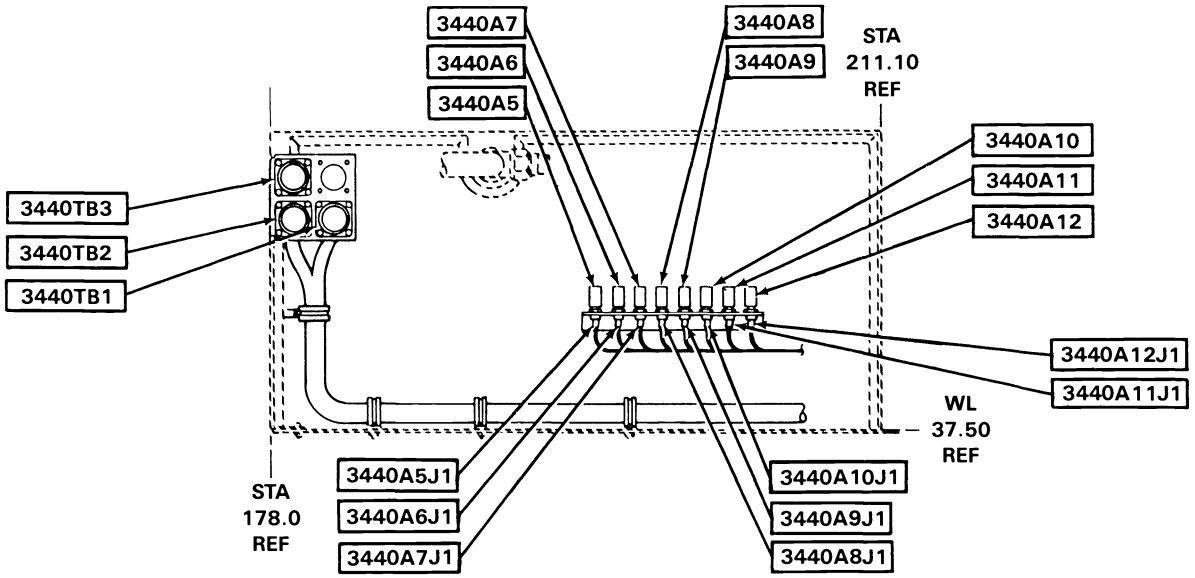
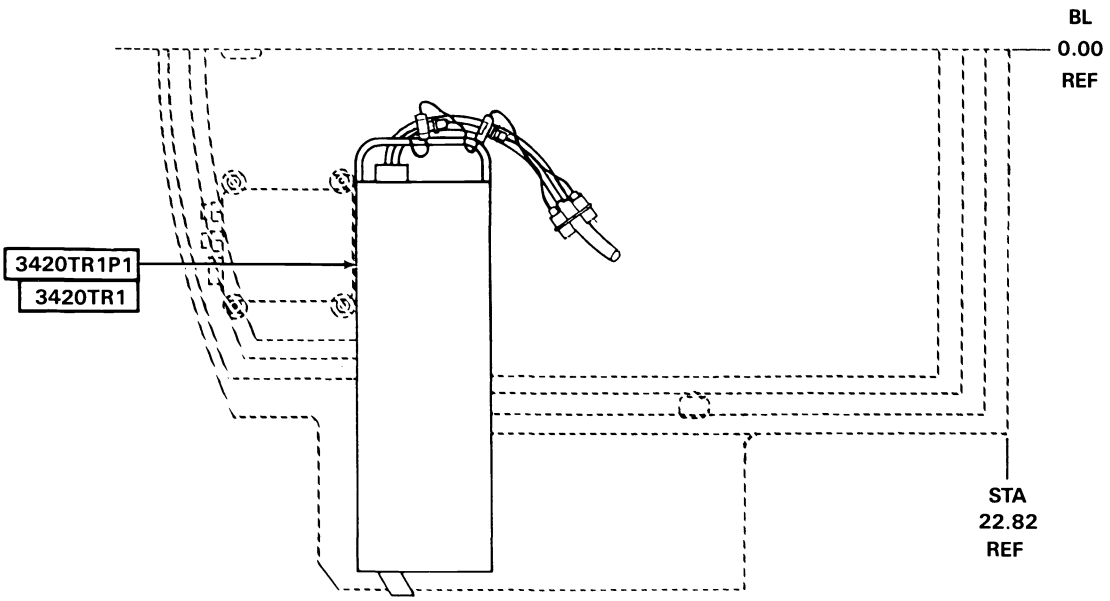



**Figure 96-4. EFIS related component location (Sheet 5)**



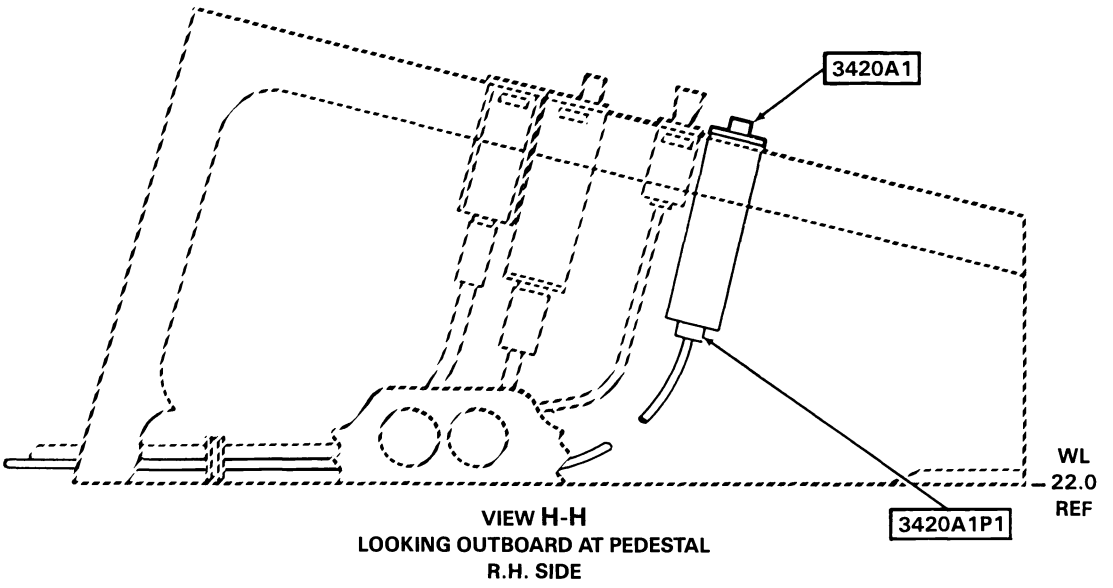

VIEW F
LOOKING INBD LH SIDE
SYMBOL GENEATORS REMOVED FOR CLARITY



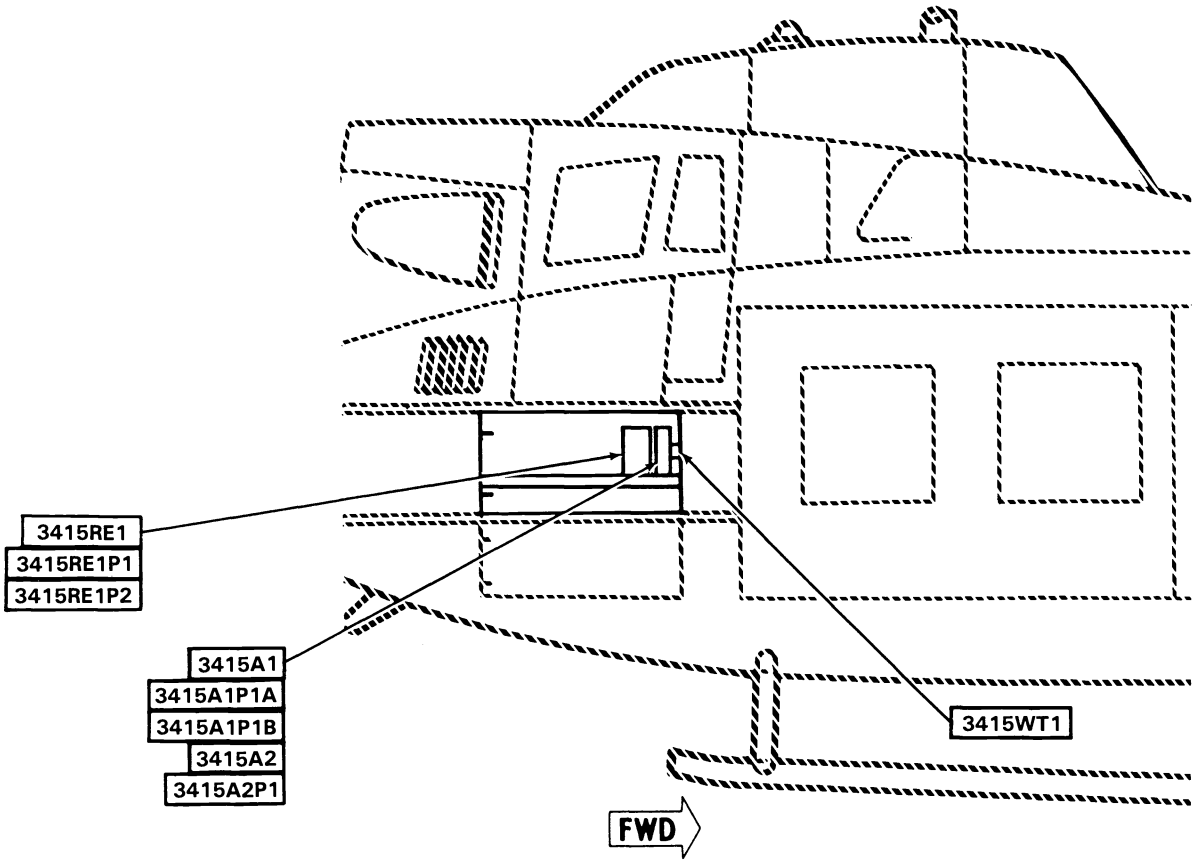

VIEW G-G
LOOKING DOWN AT LOWER NOSE
SHELF WL 22.00

412-SAR-96-4-6
412-570-002
412-705-004

**Figure 96-4. EFIS related component location (Sheet 6)**

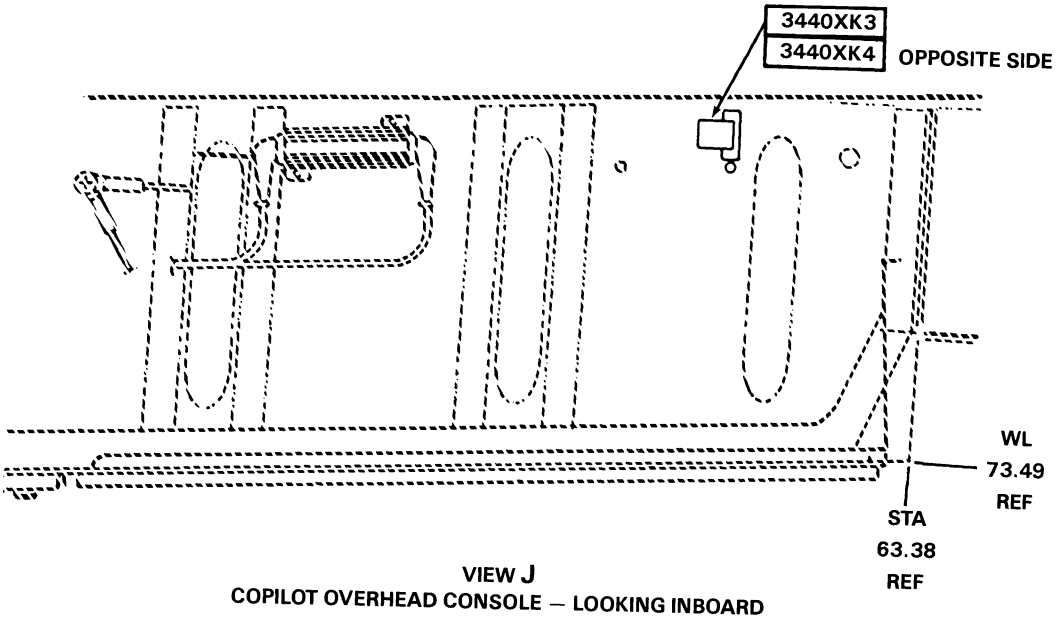



VIEW H-H
LOOKING OUTBOARD AT PEDESTAL
R.H. SIDE

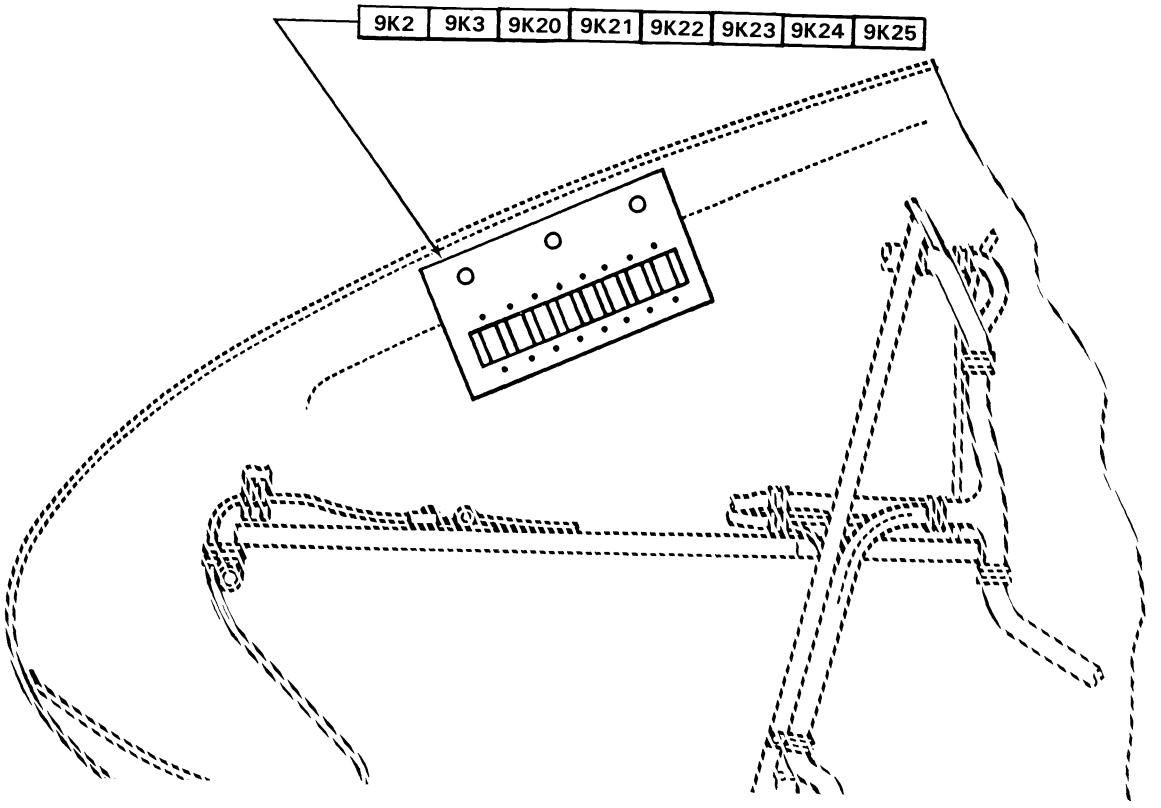



412-SAR-96-4-7

**Figure 96-4. EFIS related component location (Sheet 7)**



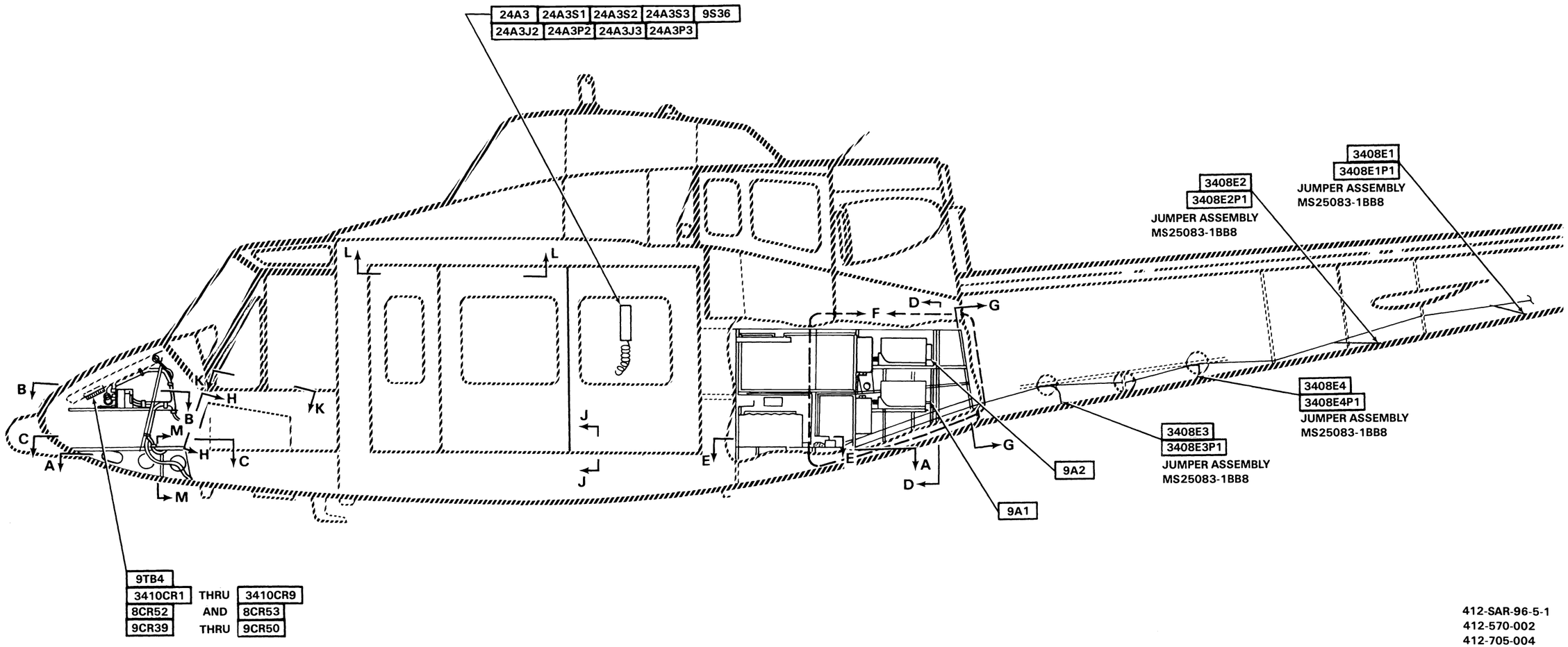

VIEW J
COPILOT OVERHEAD CONSOLE — LOOKING INBOARD




VIEW K
LOOKING OUTBOARD AT RIGHT B.L. 14.0

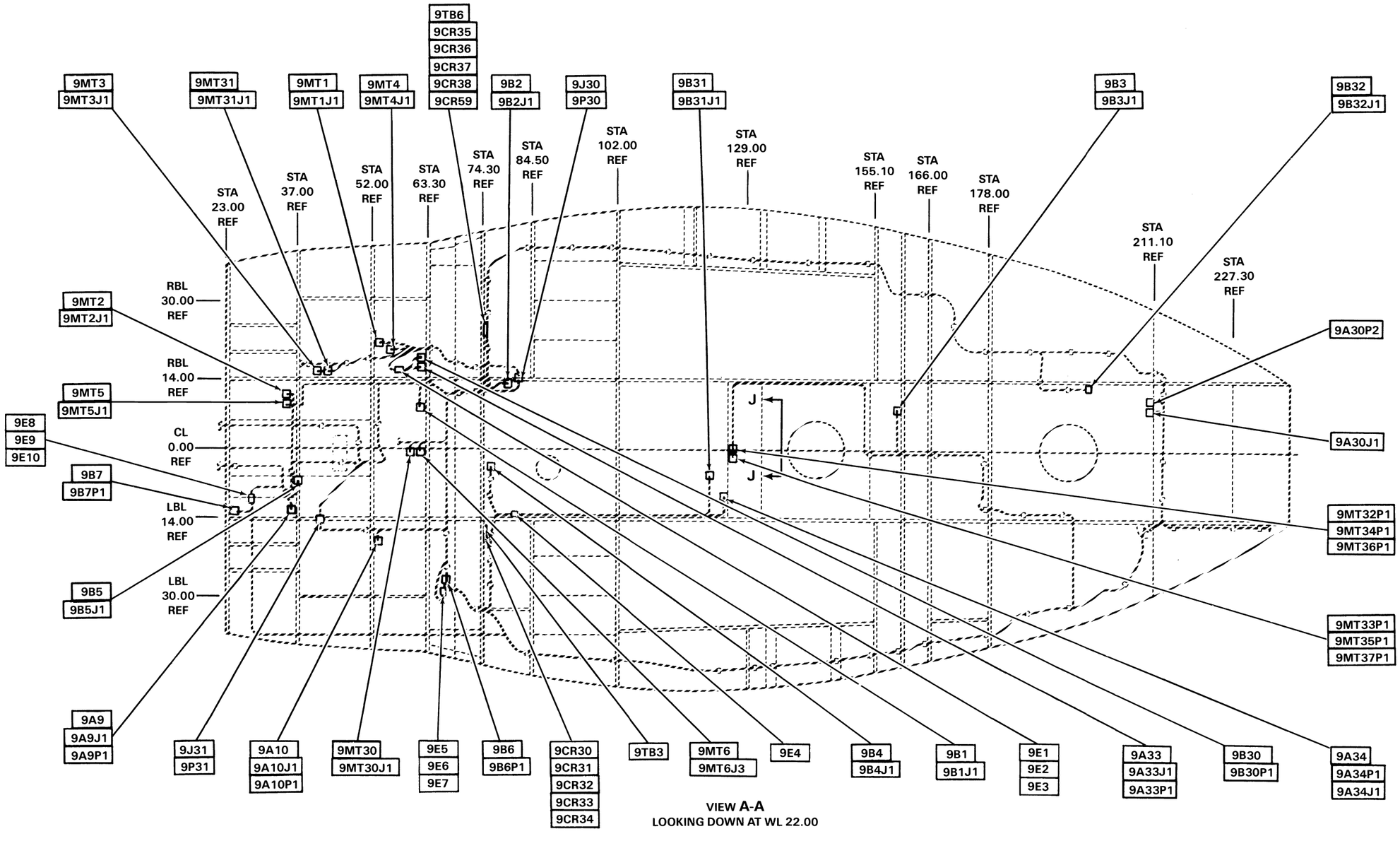
412-SAR-96-4-8

**Figure 96-4. EFIS related component location (Sheet 8)**




412-SAR-96-5-1
412-570-002
412-705-004
412-899-243

**Figure 96-5. DAFCS related component location (Sheet 1 of 11)**




VIEW **A-A**
LOOKING DOWN AT WL 22.00

412-575-007
412-SAR-96-5-2

**Figure 96-5. DAFCS related component location (Sheet 2)**

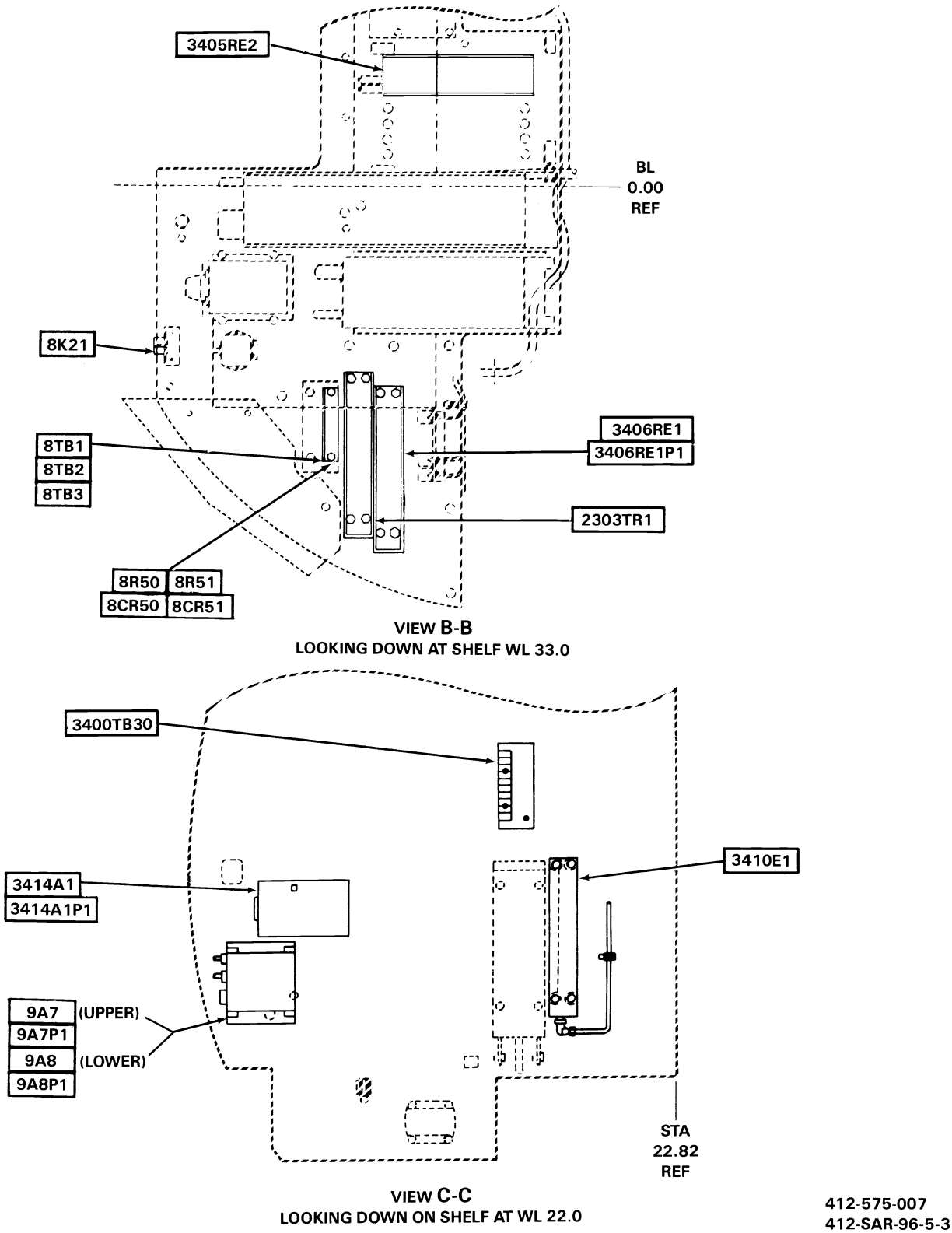
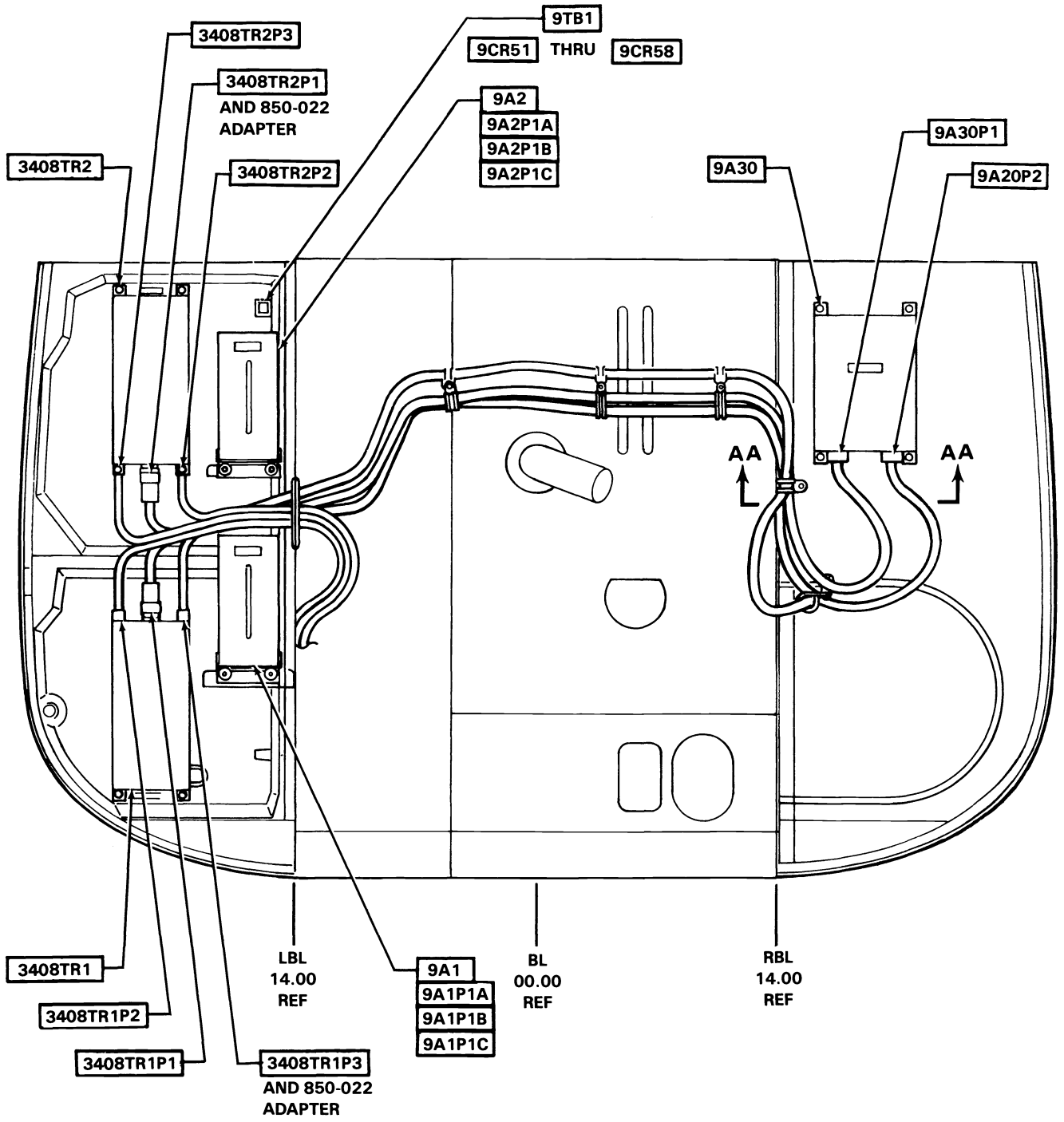


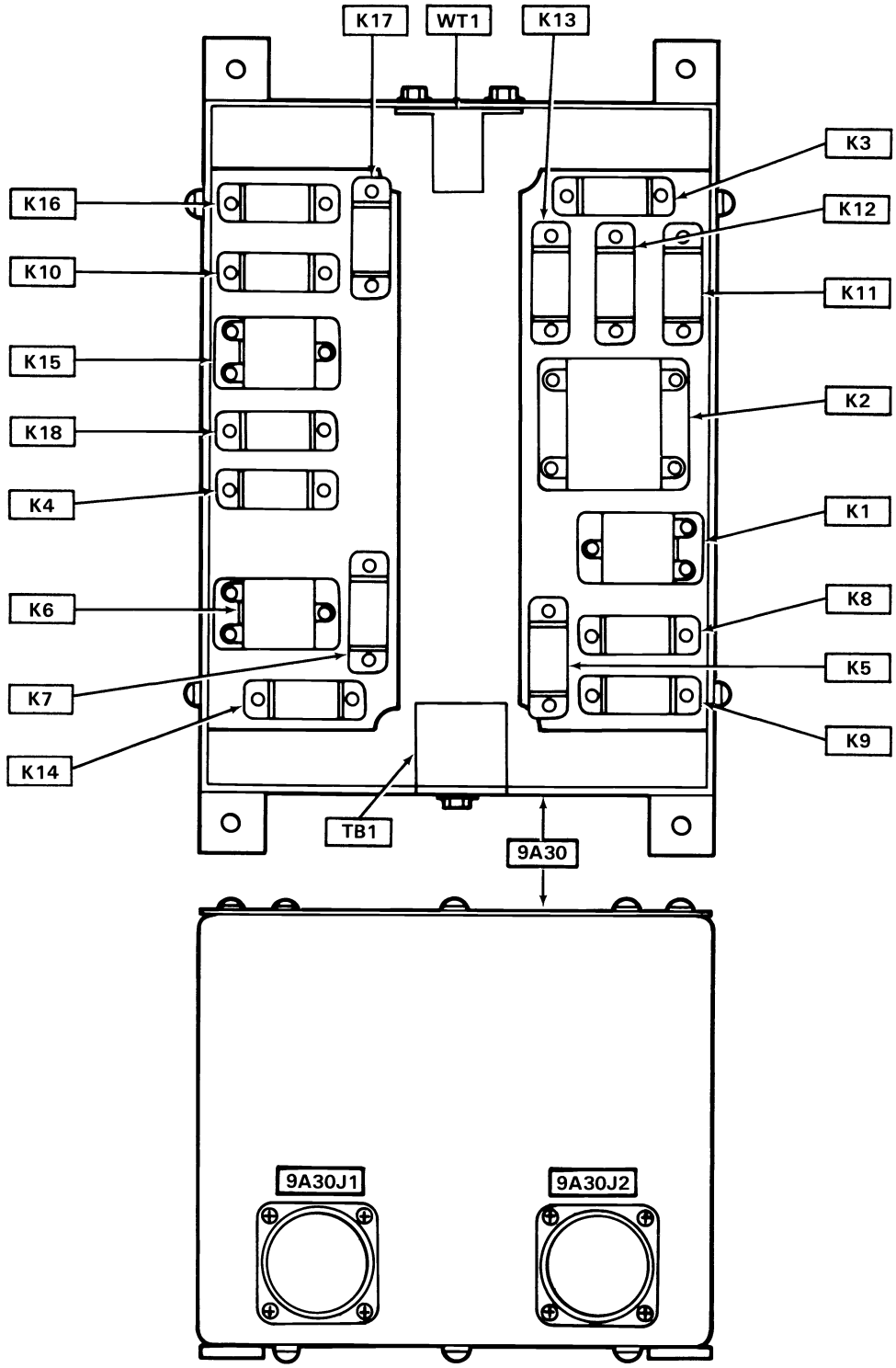

**Figure 96-5. DAFCS related component location (Sheet 3)**




VIEW D-D
LOOKING FORWARD AT STA 211.1

412-575-007
412-SAR-96-5-4

**Figure 96-5. DAFCS related component location (Sheet 4)**




VIEW AA-AA

412-SAR-96-5-5

**Figure 96-5. DAFCS related component location (Sheet 5)**

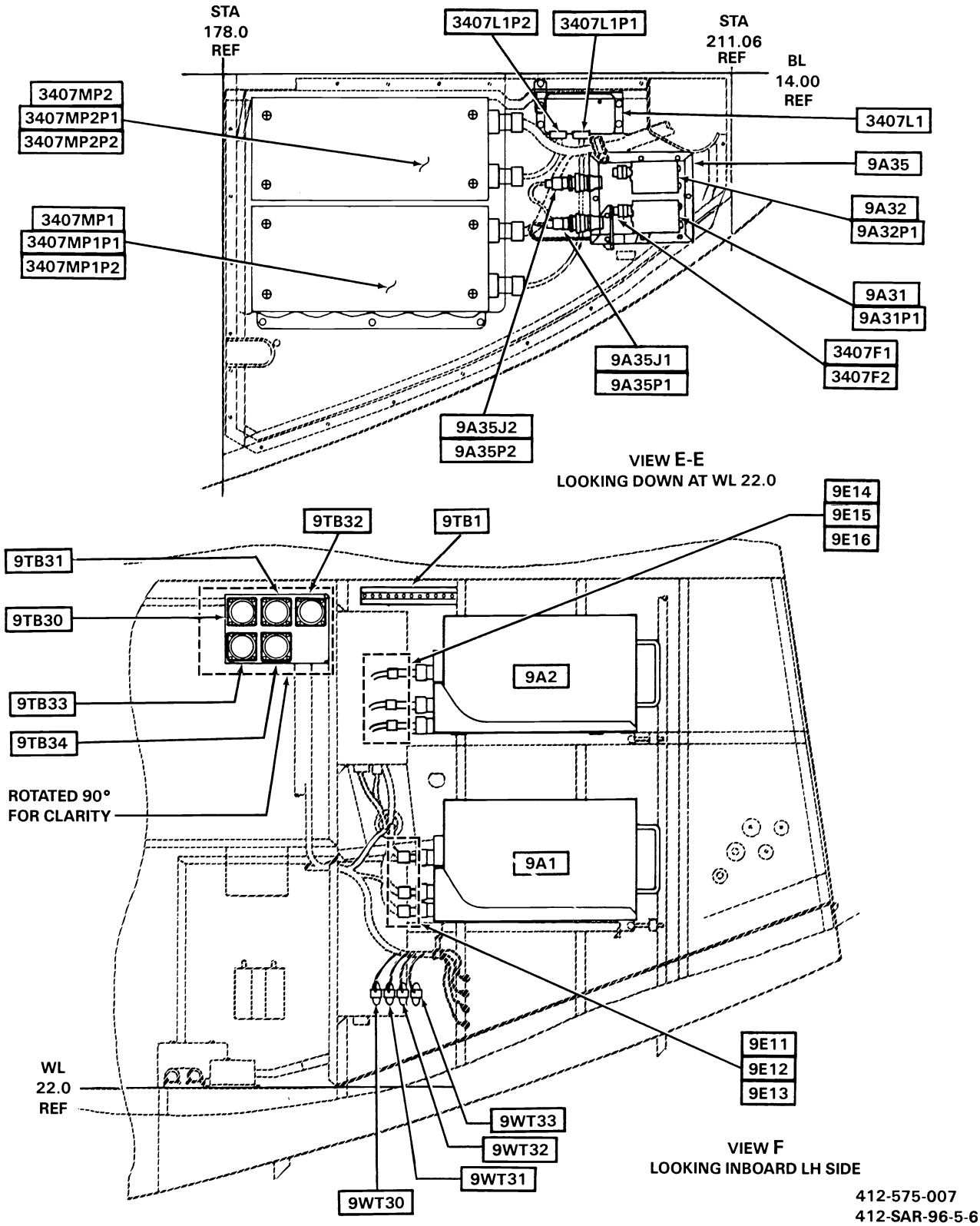
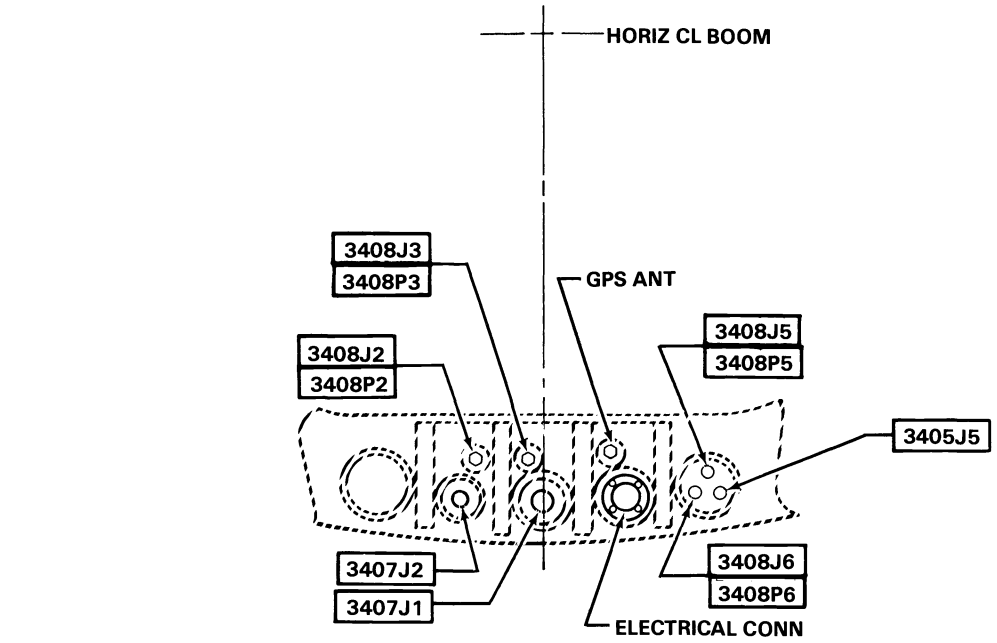
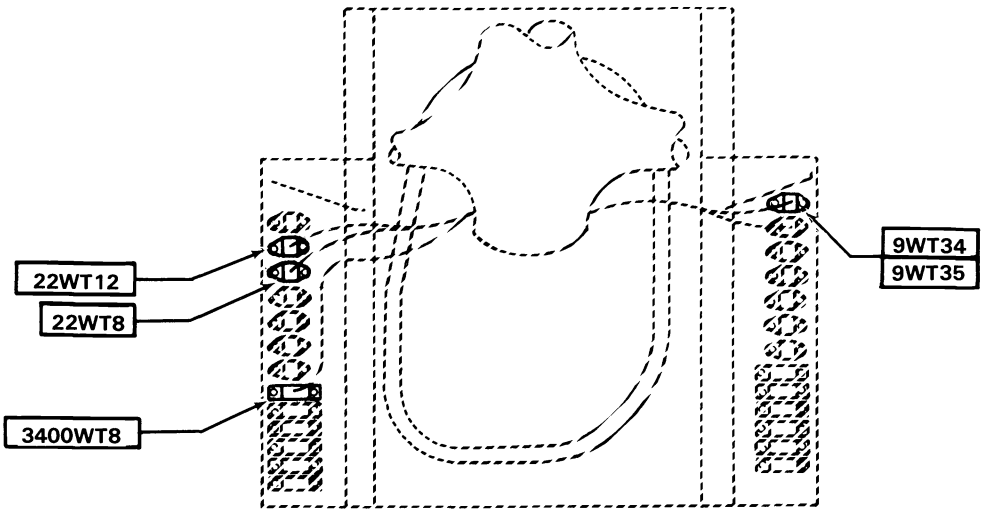



**Figure 96-5. DAFCS related component location (Sheet 6)**



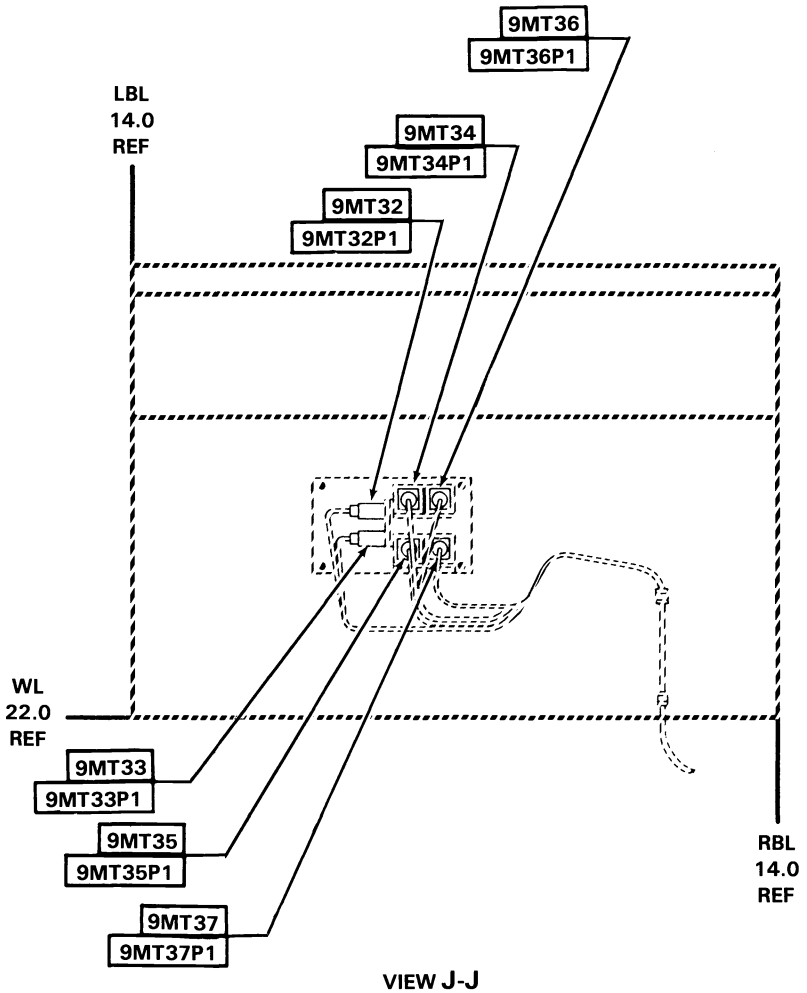

VIEW G-G
LOOKING AFT AT STA 227.3




VIEW H-H
LOOKING AFT AT FORWARD SIDE
OF PEDESTAL ROTATED 14° COUNTERCLOCKWISE

412-SAR-96-5-7
412-575-007

**Figure 96-5. DAFCS related component location (Sheet 7)**




412-575-007
412-SAR-96-5-8

**Figure 96-5. DAFCS related component location (Sheet 8)**

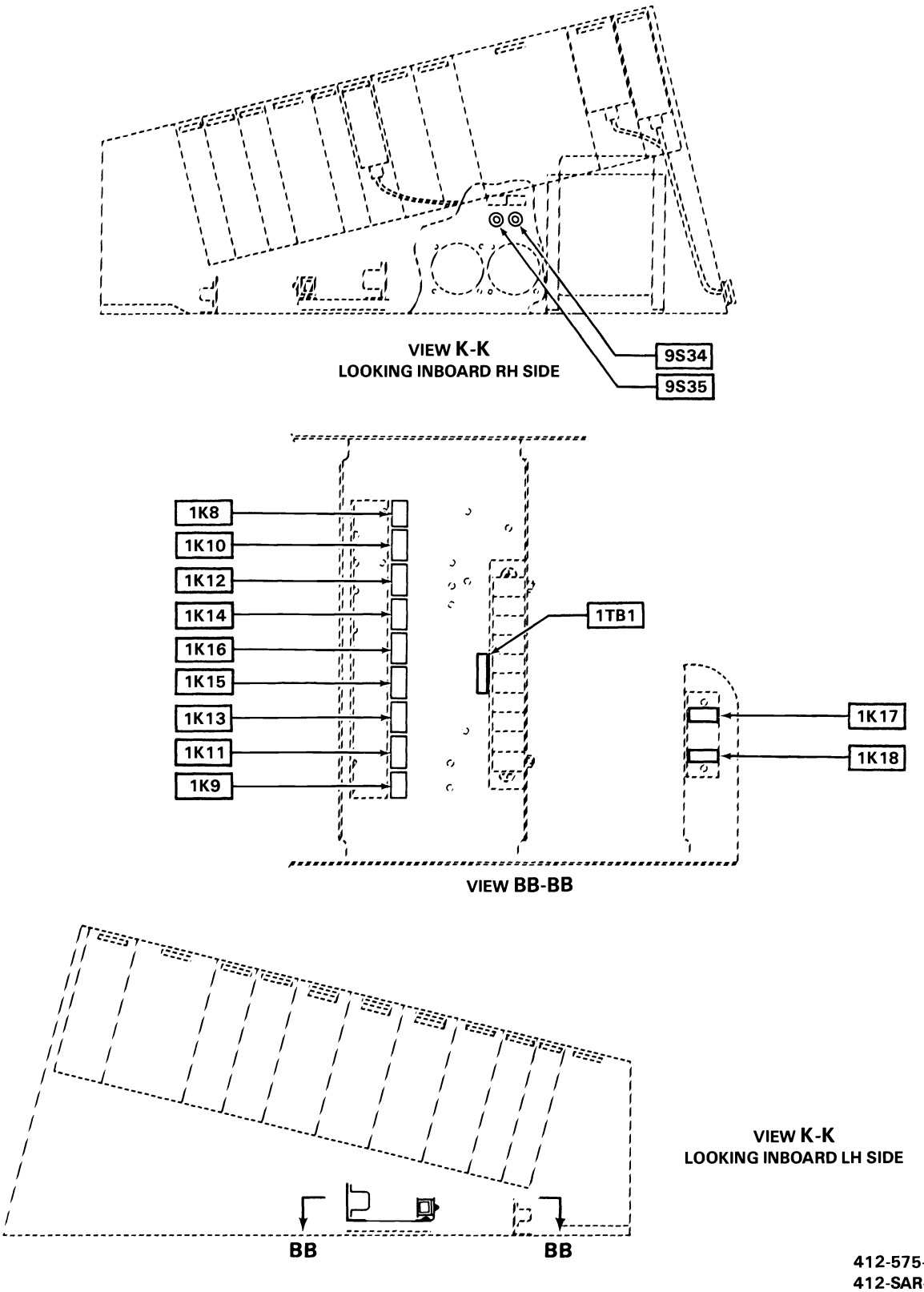



**Figure 96-5. DAFCS related component location (Sheet 9)**

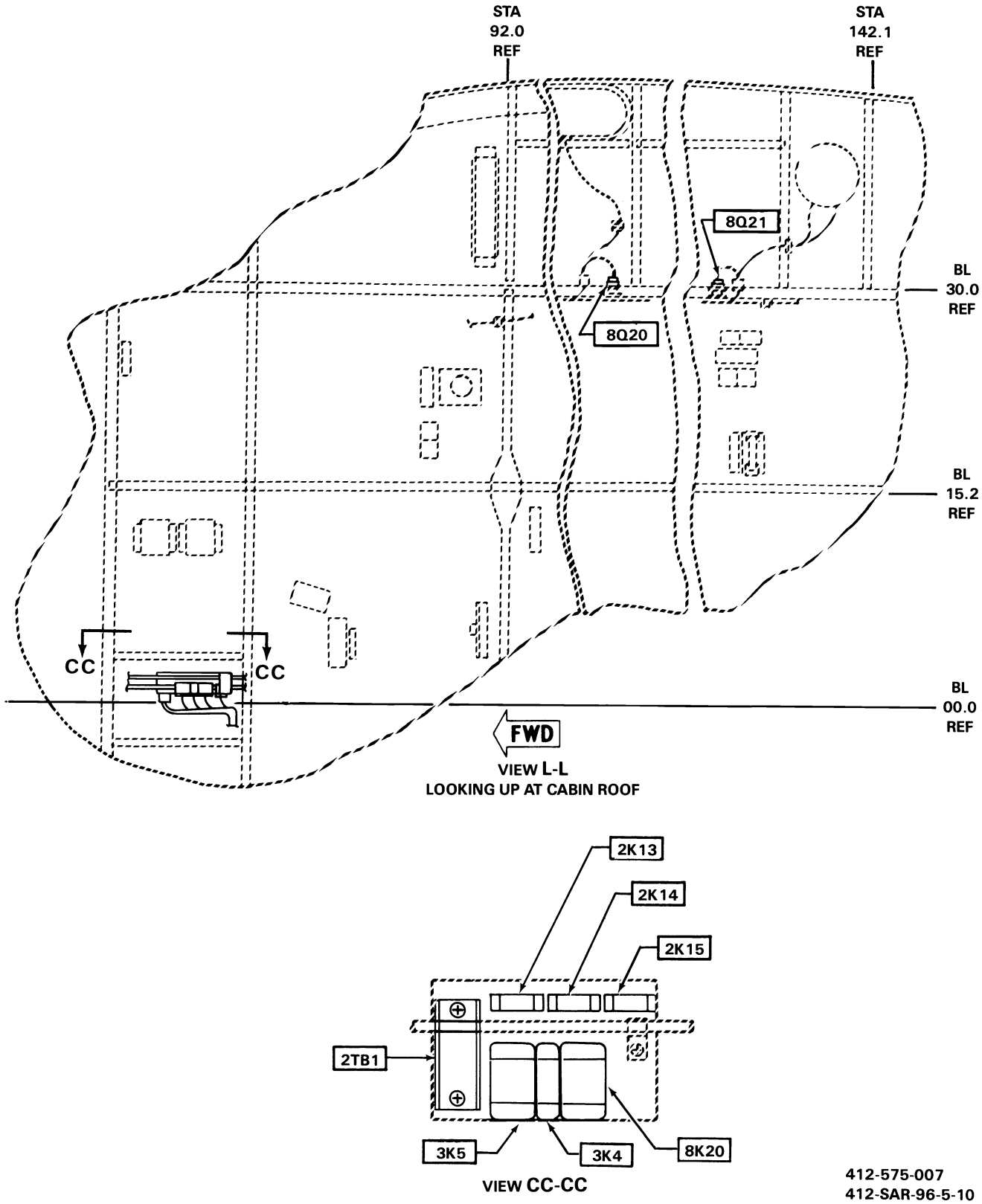
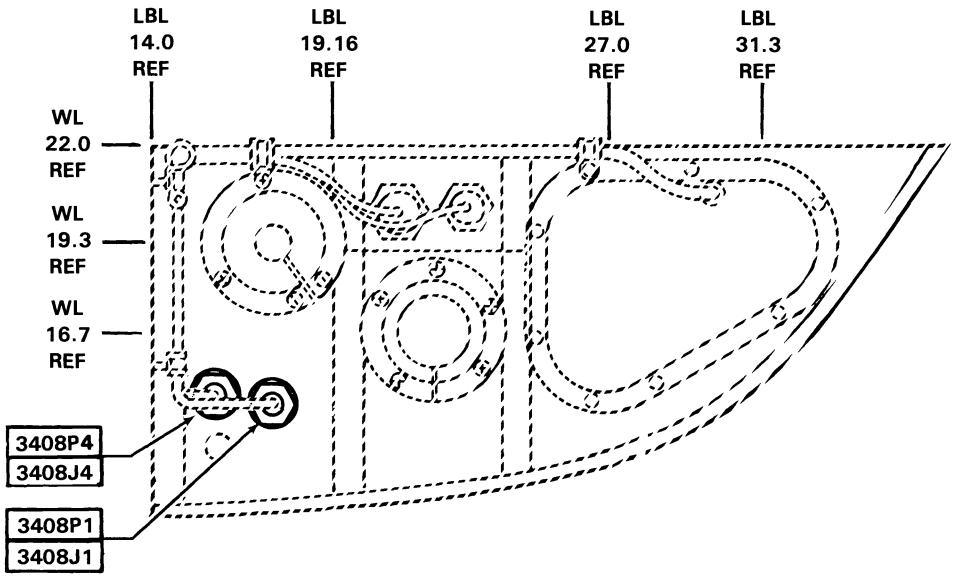



**Figure 96-5. DAFCS related component location (Sheet 10)**




VIEW M-M
LOOKING AFT AT STA 23.0
ROTATED 90° COUNTERCLOCKWISE

412-575-007
412-SAR-96-5-11

**Figure 96-5. DAFCS related component location (Sheet 11)**

**Table 96-1. Electrical equipment list**

| REF DES | NOMENCLATURE |
|---|---|
| 10DS1 | LIGHT, WINDSHIELD HEATER |
| 10DS1P1 | CONN, WINDSHIELD HEATER LIGHT |
| 22WT12 | TERMINAL BOARD, GROUND |
| 22WT14 | TERMINAL BOARD, GROUND |
| 22WT8 | TERMINAL BOARD, GROUND |
| 2300J1 | CONN |
| 24A1 | CONTROL PANEL, MISCELLANEOUS |
| 24A1P1 | CONN, MISCELLANEOUS CONTROL PANEL |
| 24A3 | CONTROL UNIT, HOIST PENDANT |
| 24A3J2 | CONN, HOIST PENDANT CONTROL |
| 24A3J3 | CONN, HOIST PENDANT CONTROL |
| 24A3P2 | CONN, HOIST PENDANT CONTROL |
| 24A3P3 | CONN, HOIST PENDANT CONTROL |
| 24A3S1 | SWITCH, BOOM |
| 24A3S3 | SWITCH, HOIST UP-DOWN |
| 24P1 | CONN, HOIST REMOTE BEEP - CABIN ROOF DISCONNECT |
| 27A2 | COMPARATOR, AIR DATA |
| 27A2P1 | CONNECTOR, AIR DATA COMPARATOR |
| 27TB1 | TERMINAL BOARD, FLIGHT DATA RECORDER |
| 3400CB3 | CIRCUIT BREAKER, HSI |
| 3400CP30 | NAV CONTROL |
| 3400CP30P1 | CONN, NAV CONTROL |
| 3400P13B | CONN |
| 3400TB30 | TERMINAL BOARD |
| 3400WT2 | TERMINAL BOARD, GROUND |
| 3400WT8 | TERMINAL BOARD, GROUND |
| 3407CP1 | PANEL, COPILOT COMPASS CONTROL |
| 3407CP1P1 | CONN, COPILOT COMPASS CONTROL PANEL |

**Table 96-1. Electrical equipment list (Cont)**

| REF DES | NOMENCLATURE |
|---|---|
| 3407CP2 | PANEL, PILOT COMPASS CONTROL |
| 3407CP2P1 | CONN, PILOT COMPASS CONTROL PANEL |
| 3407F1 | FUSE, VERTICAL GYRO |
| 3407F2 | FUSE, VERTICAL GYRO |
| 3407L1 | COMPENSATOR, DUAL REMOTE |
| 3407L1P1 | CONN, DUAL REMOTE COMPENSATOR |
| 3407L1P2 | CONN, DUAL REMOTE COMPENSATOR |
| 3407MP1 | SENSOR, COPILOT 3-AXIS REFERENCE |
| 3407MP1P1 | CONN, COPILOT 3-AXIS REFERENCE SENSOR |
| 3407MP1P2 | CONN, COPILOT 3-AXIS REFERENCE SENSOR |
| 3407MP2 | SENSOR, PILOT 3-AXIS REFERENCE |
| 3407MP2P1 | CONN, PILOT 3-AXIS REFERENCE SENSOR |
| 3407MP2P2 | CONN, PILOT 3-AXIS REFERENCE SENSOR |
| 3440A1 | SYMBOL GENERATOR, COPILOT |
| 3440A10 | FAN, EFIS |
| 3440A10 | TERMINATOR, RESISTIVE LOAD |
| 3440A10J1 | CONN, RESISTIVE LOAD TERMINATOR |
| 3440A11 | FAN, EFIS |
| 3440A11 | TERMINATOR, RESISTIVE LOAD |
| 3440A11J1 | CONN, RESISTIVE LOAD TERMINATOR |
| 3440A12 | MOD, COMM |
| 3440A12 | TERMINATOR, RESISTIVE LOAD |
| 3440A12J1 | CONN, RESISTIVE LOAD TERMINATOR |
| 3440A13 | MOD, COMM |
| 3440A1P1A | CONN, COPILOT SYMBOL GENERATOR |
| 3440A1P1B | CONN, COPILOT SYMBOL GENERATOR |

**Table 96-1. Electrical equipment list (Cont)**

| REF DES | NOMENCLATURE |
|---|---|
| 3440A1P2A | CONN, COPILOT SYMBOL GENERATOR |
| 3440A1P2B | CONN, COPILOT SYMBOL GENERATOR |
| 3440A2 | SYMBOL GENERATOR, PILOT |
| 3440A2P1A | CONN, PILOT SYMBOL GENERATOR |
| 3440A2P1B | CONN, PILOT SYMBOL GENERATOR |
| 3440A2P2A | CONN, PILOT SYMBOL GENERATOR |
| 3440A2P2B | CONN, PILOT SYMBOL GENERATOR |
| 3440A3 | SWITCH PANEL, COPILOT EFIS AUXILIARY |
| 3440A3P1 | CONN, COPILOT EFIS AUXILIARY SWITCH PANEL |
| 3440A4 | SWITCH PANEL, PILOT EFIS AUXILIARY |
| 3440A4P1 | CONN, PILOT EFIS AUXILIARY SWITCH PANEL |
| 3440A5 | TERMINATOR, RESISTIVE LOAD |
| 3440A5J1 | CONN, RESISTIVE LOAD TERMINATOR |
| 3440A6 | TERMINATOR, RESISTIVE LOAD |
| 3440A6J1 | CONN, RESISTIVE LOAD TERMINATOR |
| 3440A7 | TERMINAL, RESISTIVE LOAD |
| 3440A7J1 | CONN, RESISTIVE LOAD TERMINATOR |
| 3440A8 | TERMINATOR, RESISTIVE LOAD |
| 3440A8J1 | CONN, RESISTIVE LOAD TERMINATOR |
| 3440A9 | TERMINATOR, RESISTIVE LOAD |
| 3440A9J1 | CONN, RESISTIVE LOAD TERMINATOR |
| 3440CB1 | CIRCUIT BREAKER, EFIS NO. 1 ADI |
| 3440CB10 | CIRCUIT BREAKER, EFIS BLOWER |

**Table 96-1. Electrical equipment list (Cont)**

| REF DES | NOMENCLATURE |
|---|---|
| 3440CB2 | CIRCUIT BREAKER, EFIS NO. 2 ADI |
| 3440CB3 | CIRCUIT BREAKER, EFIS NO. 1 HSI |
| 3440CB4 | CIRCUIT BREAKER, EFIS NO. 2 HSI |
| 3440CB5 | CIRCUIT BREAKER, EFIS NO. 1 SG/DC |
| 3440CB6 | CIRCUIT BREAKER, EFIS NO. 2 SG/DC |
| 3440CB7 | CIRCUIT BREAKER, EFIS 26V NO. 1 |
| 3440CB8 | CIRCUIT BREAKER, EFIS 26V NO. 2 |
| 3440CB9 | CIRCUIT BREAKER, EFIS BLOWER |
| 3440CP1 | INSTRUMENT CONTROLLER |
| 3440CP1P1 | CONN, INSTRUMENT CONTROLLER |
| 3440CP1P2 | CONN, INSTRUMENT CONTROLLER |
| 3440CP3 | DISPLAY CONTROLLER, COPILOT |
| 3440CP3P1 | CONN, COPILOT DISPLAY CONTROLLER |
| 3440CP4 | DISPLAY CONTROLLER, PILOT |
| 3440CP4P1 | CONN, PILOT DISPLAY CONTROLLER |
| 3440CR1 | DIODE |
| 3440DS1 | EADI, COPILOT |
| 3440DS1P1 | CONN, COPILOT EADI |
| 3440DS10 | SWITCH, EFIS MASTER |
| 3440DS2 | PILOT, EADI |
| 3440DS2P1 | CONN, PILOT EADI |
| 3440DS3 | EHSI, COPILOT |
| 3440DS3P1 | CONN, COPILOT EHSI |
| 3440DS4 | EHSI, PILOT |
| 3440DS4P1 | CONN, PILOT EHSI |
| 3440J1 | CONN |
| 3440J10 | CONN, TEE |
| 3440J10C | CONN |
| 3440J11 | CONN, TEE |
| 3440J11C | CONN |
| 3440J12 | CONN, TEE |
| 3440J12C | CONN |
| 3440J13 | CONN, TEE |

**Table 96-1. Electrical equipment list (Cont)**

| REF DES | NOMENCLATURE |
|---|---|
| 3440J13C | CONN |
| 3440J14 | CONN, TEE |
| 3440J14C | CONN |
| 3440J15 | CONN, TEE |
| 3440J15C | CONN |
| 3440J16 | CONN, TEE |
| 3440J16C | CONN |
| 3440J17 | CONN, TEE |
| 3440J17C | CONN |
| 3440J2 | CONN |
| 3440J3 | CONN |
| 3440J4 | CONN |
| 3440K1 | RELAY |
| 3440K2 | RELAY |
| 3440K3 | RELAY |
| 3440K4 | RELAY |
| 3440K5 | RELAY |
| 3440K6 | RELAY |
| 3440P1 | CONN |
| 3440P10A | CONN |
| 3440P10B | CONN |
| 3440P11A | CONN |
| 3440P11B | CONN |
| 3440P12A | CONN |
| 3440P12B | CONN |
| 3440P13A | CONN |
| 3440P14A | CONN |
| 3440P14B | CONN |
| 3440P15A | CONN |
| 3440P15B | CONN |
| 3440P16A | CONN |
| 3440P16B | CONN |
| 3440P17A | CONN |
| 3440P17B | CONN |
| 3440P2 | CONN |
| 3440P3 | CONN |
| 3440P4 | CONN |
| 3440TB1 | TERMINAL BOARD |
| 3440TB2 | TERMINAL BOARD |
| 3440TB3 | TERMINAL BOARD |
| 3440WT1 | TERMINAL BOARD, GROUND |
| 3440WT2 | TERMINAL BOARD, GROUND |
| 4A2 | COLLECTIVE STICK, PILOT |
| 4A2J1 | CONN, PILOT COLLECTIVE STICK |

**Table 96-1. Electrical equipment list (Cont)**

| REF DES | NOMENCLATURE |
|---|---|
| 4M5 | CLOCK, PILOT DIGITAL |
| 4M5P1 | CONN, PILOT DIGITAL CLOCK |
| 5A2 | COLLECTIVE STICK, COPILOT |
| 5A2J1 | CONN, COPILOT COLLECTIVE STICK |
| 5M5 | CLOCK, COPILOT DIGITAL |
| 5M5P1 | CONN, COPILOT DIGITAL CLOCK |
| 8CR52 | DIODE |
| 8CR53 | DIODE |
| 8A1 | PANEL, CAUTION |
| 8A1P1 | CONN |
| 8A1P2 | CONN |
| 8K20 | RELAY, SWITCHING |
| 8Q20 | TRANSISTOR |
| 8Q21 | TRANSISTOR |
| 8TB1 | TERMINAL BOARD, LIGHTING |
| 8TB2 | TERMINAL BOARD, RPM WARNING |
| 8TB3 | TERMINAL BOARD, 5 VOLT LIGHT POWER |
| 9A1 | COMPUTER, COPILOT DFC NO. 1 |
| 9A10 | SWITCH, PITCH DETENT |
| 9A10J1 | CONN, PITCH DETENT SWITCH |
| 9A10P1 | CONN, PITCH DETENT SWITCH |
| 9A11 | INDICATOR PANEL AFCS ACTUATOR |
| 9A11P1 | CONN, AFCS ACTUATOR PANEL |
| 9A1P1A | CONN, COPILOT DFC COMPUTER NO. 1 |
| 9A1P1B | CONN, COPILOT DFC COMPUTER NO. 1 |
| 9A1P1C | CONN, COPILOT DFC COMPUTER NO. 1 |
| 9A2 | COMPUTER, PILOT DFC NO. 2 |
| 9A2P1A | CONN, PILOT DFC COMPUTER NO. 2 |
| 9A2P1B | CONN, PILOT DFC COMPUTER NO. 2 |

**Table 96-1. Electrical equipment list (Cont)**

| REF DES | NOMENCLATURE |
|---|---|
| 9A2P1C | CONN, PILOT DFC COMPUTER NO. 2 |
| 9A3 | MODE SELECTOR, COPILOT |
| 9A3P1 | CONN, COPILOT MODE SELECTOR |
| 9A30 | CONTROL BOX, AFCS RELAY |
| 9A30P1 | CONN, AFCS RELAY CONTROL BOX |
| 9A30P1 | CONN, AFCS RELAY CONTROL BOX |
| 9A31 | GYRO NO. 1, YAW RATE |
| 9A31P1 | CONN, YAW RATE GYRO NO. 1 |
| 9A32 | GYRO NO. 2, YAW RATE |
| 9A32P1 | CONN, YAW RATE GYRO NO. 2 |
| 9A33 | SWITCH, YAW DETENT |
| 9A33J1 | CONN, YAW DETENT SWITCH |
| 9A33P1 | CONN, YAW DETENT SWITCH |
| 9A34 | SWITCH, COLLECTIVE DETENT |
| 9A34J1 | CONN, COLLECTIVE DETENT SWITCH |
| 9A34P1 | CONN, COLLECTIVE DETENT SWITCH |
| 9A35 | GYRO ASSEMBLY, YAW RATE |
| 9A35J1 | CONN, YAW RATE GYRO ASSEMBLY |
| 9A35J2 | CONN, YAW RATE GYRO ASSEMBLY |
| 9A35P1 | CONN, YAW RATE GYRO ASSEMBLY |
| 9A35P2 | CONN, YAW RATE GYRO ASSEMBLY |
| 9A3P1 | CONN, COPILOT MODE SELECTOR |
| 9A3P1 | CONN, COPILOT MODE SELECTOR |
| 9A4 | MODE SELECTOR, PILOT |
| 9A4P1 | CONN, PILOT MODE SELECTOR |
| 9A5 | CONTROLLER, AUTOPILOT |

**Table 96-1. Electrical equipment list (Cont)**

| REF DES | NOMENCLATURE |
|---|---|
| 9A5P1 | CONN, AUTOPILOT CONTROLLER |
| 9A5P2 | CONN, AUTOPILOT CONTROLLER |
| 9A6 | PRESELECT, ALTITUDE |
| 9A6P1 | CONN, ALTITUDE PRESELECT |
| 9A7 | SENSOR, AIR DATA NO. 1 |
| 9A7P1 | CONN, AIR DATA SENSOR NO. 1 |
| 9A8 | SENSOR, AIR DATA NO. 2 |
| 9A8P1 | CONN, AIR DATA SENSOR NO. 2 |
| 9A9 | SWITCH, ROLL DETENT |
| 9A9J1 | CONN, ROLL DETENT SWITCH |
| 9A9P1 | CONN, ROLL DETENT SWITCH |
| 9B1 | ACTUATOR, PILOT LINEAR NO. 1 |
| 9B1J1 | CONN, PITCH LINEAR ACTUATOR NO. 1 |
| 9B2 | ACTUATOR, ROLL LINEAR NO. 1 |
| 9B2J1 | CONN, ROLL LINEAR ACTUATOR NO. 1 |
| 9B3 | ACTUATOR, YAW LINEAR NO. 1 |
| 9B30 | ACTUATOR, YAW TRIM |
| 9B30P1 | CONN, YAW TRIM ACTUATOR |
| 9B31 | ACTUATOR, COLLECTIVE TRIM |
| 9B31P1 | CONN, COLLECTIVE TRIM ACTUATOR |
| 9B32 | LINEAR ACTUATOR, YAW NO. 2 |
| 9B32J1 | CONN, YAW LINEAR ACTUATOR NO. 2 |
| 9B3J1 | CONN, YAW LINEAR ACTUATOR NO. 1 |
| 9B4 | LINEAR ACTUATOR, PITCH NO. 2 |
| 9B4J1 | CONN, PITCH LINEAR ACTUATOR NO. |
| 9B5 | LINEAR ACTUATOR, ROLL NO. 2 |

**Table 96-1. Electrical equipment list (Cont)**

| REF DES | NOMENCLATURE |
|---|---|
| 9B5J1 | CONN, ROLL LINEAR ACTUATOR NO. 2 |
| 9B6 | ACTUATOR, PITCH TRIM |
| 9B6P1 | CONN, PITCH TRIM ACTUATOR |
| 9B7 | ACTUATOR, YAW TRIM |
| 9B7P1 | CONN, YAW TRIM ACTUATOR |
| 9CB2 | CIRCUIT BREAKER, AFCS NO. 2 |
| 9CB3 | CIRCUIT BREAKER, FORCE TRIM |
| 9CB30 | CIRCUIT BREAKER, RATE GYRO NO. 2 |
| 9CB31 | CIRCUIT BREAKER, RATE GYRO NO. 1 |
| 9CB32 | CIRCUIT BREAKER, AFCS NO. 1 ADD/CAUTION |
| 9CB33 | CIRCUIT BREAKER, AFCS NO. 2 CAUT |
| 9CB4 | CIRCUIT BREAKER, AFCS 26V NO. 1 |
| 9CB7 | CIRCUIT BREAKER, AFCS NO. 2 |
| 9CR30 | DIODE |
| 9CR31 | DIODE |
| 9CR32 | DIODE |
| 9CR33 | DIODE |
| 9CR34 | DIODE |
| 9CR35 | DIODE |
| 9CR36 | DIODE |
| 9CR37 | DIODE |
| 9CR38 | DIODE |
| 9CR39 | DIODE |
| 9CR40 | DIODE |
| 9CR41 | DIODE |
| 9CR42 | DIODE |
| 9CR43 | DIODE |
| 9CR44 | DIODE |
| 9CR45 | DIODE |
| 9CR46 | DIODE |
| 9CR47 | DIODE |
| 9CR48 | DIODE |
| 9CR49 | DIODE |
| 9CR50 | DIODE |
| 9CR51 | DIODE |
| 9CR52 | DIODE |

**Table 96-1. Electrical equipment list (Cont)**

| REF DES | NOMENCLATURE |
|---|---|
| 9CR53 | DIODE |
| 9CR54 | DIODE |
| 9CR55 | DIODE |
| 9CR56 | DIODE |
| 9CR57 | DIODE |
| 9CR58 | DIODE |
| 9CR59 | DIODE |
| 9DS30 | LIGHT, COPILOT AUTOPILOT |
| 9DS30P1 | CONN, COPILOT AUTOPILOT LIGHT |
| 9DS31 | LIGHT COPILOT PITCH, ROLL, YAW |
| 9DS31P1 | CONN, COPILOT PITCH, ROLL, YAW LIGHT |
| 9DS32 | LIGHT, COPILOT FORCE TRIM DECOUPLER |
| 9DS32P1 | CONN, COPILOT FORCE TRIM DECOUPLER LIGHT |
| 9DS33 | LIGHT, PILOT AUTOPILOT |
| 9DS33P1 | CONN, PILOT AUTOPILOT LIGHT |
| 9DS34 | LIGHT, PILOT PITCH, ROLL, YAW |
| 9DS34P1 | CONN, PILOT PITCH, ROLL, YAW LIGHT |
| 9DS35 | LIGHT, PILOT FORCE TRIM DECOUPLER |
| 9DS35P1 | CONN, PILOT FORCE TRIM DECOUPLER LIGHT |
| 9DS36 | LIGHT, MOT/ON |
| 9DS36P1 | CONN, MOT/ON LIGHT |
| 9DS37 | LIGHT, CLIMB/ON |
| 9DS37P1 | CONN, CLIMB/ON LIGHT |
| 9DS38 | LIGHT, APR/1/2 |
| 9DS38P1 | CONN, APR/1/2 LIGHT |
| 9DS39 | LIGHT, HOIST/EN |
| 9DS39P1 | CONN, HOIST/EN LIGHT |
| 9J30 | CONN |
| 9J31 | CONN |
| 9K2 | RELAY, TRIM FAIL ANNUN |
| 9K20 | RELAY |
| 9K21 | RELAY |
| 9K22 | RELAY |
| 9K23 | RELAY |
| 9K24 | RELAY |
| 9K25 | RELAY |

Table 96-1. Electrical equipment list (Cont)

| REF DES | NOMENCLATURE |
|---|---|
| 9K3 | RELAY, CLTV WARN ANNUN |
| 9MT1 | DUAL CMT, COPILOT CYCLIC PITCH |
| 9MT1J1 | CONN, COPILOT CYCLIC PITCH DUAL CMT |
| 9MT2 | DUAL CMT, COPILOT CYCLIC ROLL |
| 9MT2J1 | CONN, COPILOT CYCLIC ROLL DUAL CMT |
| 9MT3 | DUAL CMT, COPILOT PEDALS YAW |
| 9MT30 | DUAL CMT, PILOT COLLECTIVE |
| 9MT30J1 | CONN, PILOT COLLECTIVE DUAL CMT |
| 9MT31 | DUAL CMT, PILOT PEDALS YAW |
| 9MT31J1 | CONN, PILOT PEDALS YAW DUAL CMT |
| 9MT32 | ACCELEROMETER, LONGITUDE NO. 1 |
| 9MT32P1 | CONN, LONGITUDE NO. 1 ACCELEROMETER |
| 9MT33 | ACCELEROMETER, LONGITUDE |
| 9MT33P1 | CONN, LONGITUDE ACCELEROMETER |
| 9MT34 | ACCELEROMETER, LATITUDE NO. 1 |
| 9MT34P1 | CONN, LATITUDE NO. 1 ACCELEROMETER |
| 9MT35 | ACCELEROMETER, LATITUDE |
| 9MT35P1 | CONN, LATITUDE ACCELEROMETER |
| 9MT36 | ACCELEROMETER, VERTICAL NO. 1 |
| 9MT36P1 | CONN, VERTICAL NO. 1 ACCELEROMETER |
| 9MT37 | ACCELEROMETER, VERTICAL |
| 9MT37P1 | CONN, VERTICAL ACCELEROMETER |
| 9MT3J1 | CONN, COPILOT PEDALS YAW DUAL CMT |

Table 96-1. Electrical equipment list (Cont)

| REF DES | NOMENCLATURE |
|---|---|
| 9MT4 | DUAL CMT, PILOT PITCH CYCLIC |
| 9MT4J1 | CONN, PILOT PITCH CYCLIC DUAL CMT |
| 9MT5 | DUAL CMT, PILOT ROLL CYCLIC |
| 9MT5J1 | CONN, PILOT ROLL CYCLIC DUAL CMT |
| 9MT6 | DUAL CMT, COPILOT COLLECTIVE |
| 9MT6J1 | CONN, COPILOT COLLECTIVE DUAL CMT |
| 9P30 | CONN |
| 9P31 | CONN |
| 9S34 | SWITCH, READ EEPROM |
| 9S35 | SWITCH, ERASE EEPROM |
| 9S36 | SWITCH, HOIST REMOTE BEEP |
| 9TB1 | TERMINAL BOARD |
| 9TB3 | TERMINAL BOARD |
| 9TB30 | TERMINAL BOARD |
| 9TB31 | TERMINAL BLOCK |
| 9TB32 | TERMINAL BOARD |
| 9TB33 | TERMINAL BOARD |
| 9TB34 | TERMINAL BOARD |
| 9TB4 | TERMINAL BOARD |
| 9TB6 | TERMINAL BOARD |
| 9WT30 | TERMINAL BOARD, GROUND |
| 9WT31 | TERMINAL BOARD, GROUND |
| 9WT32 | TERMINAL BOARD, GROUND |
| 9WT33 | TERMINAL BOARD, GROUND |
| 9WT34 | TERMINAL BOARD, GROUND |
| 9WT35 | TERMINAL BOARD, GROUND |
| J14 | CONN, GLARE SHIELD FLOOD LIGHT ASSEMBLY |
| P14 | CONN, GLARE SHIELD FLOOD LIGHT ASSEMBLY |

# MISCELLANEOUS SYSTEMS

## 96-2. MISCELLANEOUS SYSTEMS.

The miscellaneous system peculiar to SAR configured helicopters is the EFIS cooling system. Refer to Chapter 98 for EFIS wiring diagrams.

## 96-3. EFIS COOLING SYSTEM.

One external cooling source is provided for each EADI/EHSI pair of electronic displays to provide adequate cooling and proper display operation. Each fan supplies a minimum of 120 cubic feet per minute free air delivery at sea level.

## 96-4. OPERATIONAL CHECK EFIS COOLING SYSTEM.

The EFIS cooling system operates when the EFIS system is operating. Refer to Honeywell, SPZ-7600 Integrated Flight Control System Bell 412HP System Maintenance Manual, Publication Number A09-1169-024.

## 96-5. REMOVAL PROCEDURES.

**WARNING**

ENSURE ELECTRICAL POWER IS OFF.

## 96-6. EFIS fan.

1. Disconnect electrical connector.
2. Remove and retain mounting hardware.
3. Remove fan.

## 96-7. INSTALLATION PROCEDURES.

**WARNING**

ENSURE ELECTRICAL POWER IS OFF.

## 96-8. EFIS fan.

1. Position fan in mounting bracket.
2. Install mounting hardware.
3. Connect electrical connector.

# INTEGRATED FLIGHT CONTROL SYSTEM

## 96-9. INTEGRATED FLIGHT CONTROL SYSTEM.

The Integrated Flight Control System consists of the Digital Automatic Flight Control System (DAFCS) and the Electronic Flight Instrument System (EFIS).

Refer to Honeywell, SPZ-7600 Integrated Flight Control System Bell 412HP System Maintenance Manual, Publication Number A09-1169-024.

## 96-10. DIGITAL AUTOMATIC FLIGHT CONTROL SYSTEM (DAFCS).

The DAFCS is comprised of the following major components: copilot and pilot digital flight control (DFC) computers (9A1 and 9A2), copilot and pilot flight director mode selectors (9A3 and 9A4), autopilot controller (9A5), attitude preselect indicator (9A6), and AFCS position indicator panel (9A11)( figure 96-6 ).

The DAFCS provides fully coupled autopilot and flight director functions in the pitch, roll, yaw, and collective axis. The autopilot and flight director functions are performed by two identical, interchangeable digital flight control computers.

The DFC computer utilizes a combination of digital and analog computational techniques to implement various automatic flight control system functions. The computer processes information about actual aircraft attitude versus a desired attitude as a function of selected flight mode to produce autopilot pitch, roll, yaw, and collective control outputs and flight director pitch and roll steering commands.

## 96-11. SAR DAFCS OPERATIONAL CHECK.

**1.** Visually inspect equipment and correct any faulty mechanical mountings or electrical connections. Connect external power, adjusted to 27.5 Vdc, to helicopter power receptacle.

**2.** Apply external hydraulic pressure, refer to BHT-412-MM , Chapter 29 .

**3.** Energize power buses per paragraph 97-4.

**4.** Close EFIS No. 1 circuit breakers ADI (3440CB1), HSI (3440CB3), SG/DL (3440CB5), EFIS 26V NO. 1 (3440CB7), BLWR (3440CB9), and EFIS No.2 circuit breakers ADI (3440CB2), HSI (3440CB4), SG/DL (3440CB6), EFIS 26V NO.2 (3440CB8), and BLWR (3440CB10).

**5.** Close circuit breakers AFCS 26V NO. 1 (9CB4), RATE GYRO NO.1 (9CB31), AFCS NO.1 PWR (9CB31), AFCS NO. 1 ADD/CAUT (9CB32), and FORCE TRIM (9CB3).

**6.** Close circuit breakers AFCS 26V NO. 2 (9CB7), RATE GYRO NO. 2 (9CB30), AFCS NO. 2 PWR (9CB2), AFCS NO.2 CAUT (9CB33), and ENC ALT (3410CB1).

**7.** Set EFIS MASTER switch to ON.

**8.** Perform operational checkout according to Honeywell, SPZ-7600 Integrated Flight Control System Bell 412HP System Maintenance Manual, Publication Number A09-1169-024.

## 96-12. TROUBLESHOOTING.

It is assumed that all appropriate circuit breakers are closed, interconnecting cabling is in normal operational condition, and power is applied to the system. (See appropriate wiring diagram.)

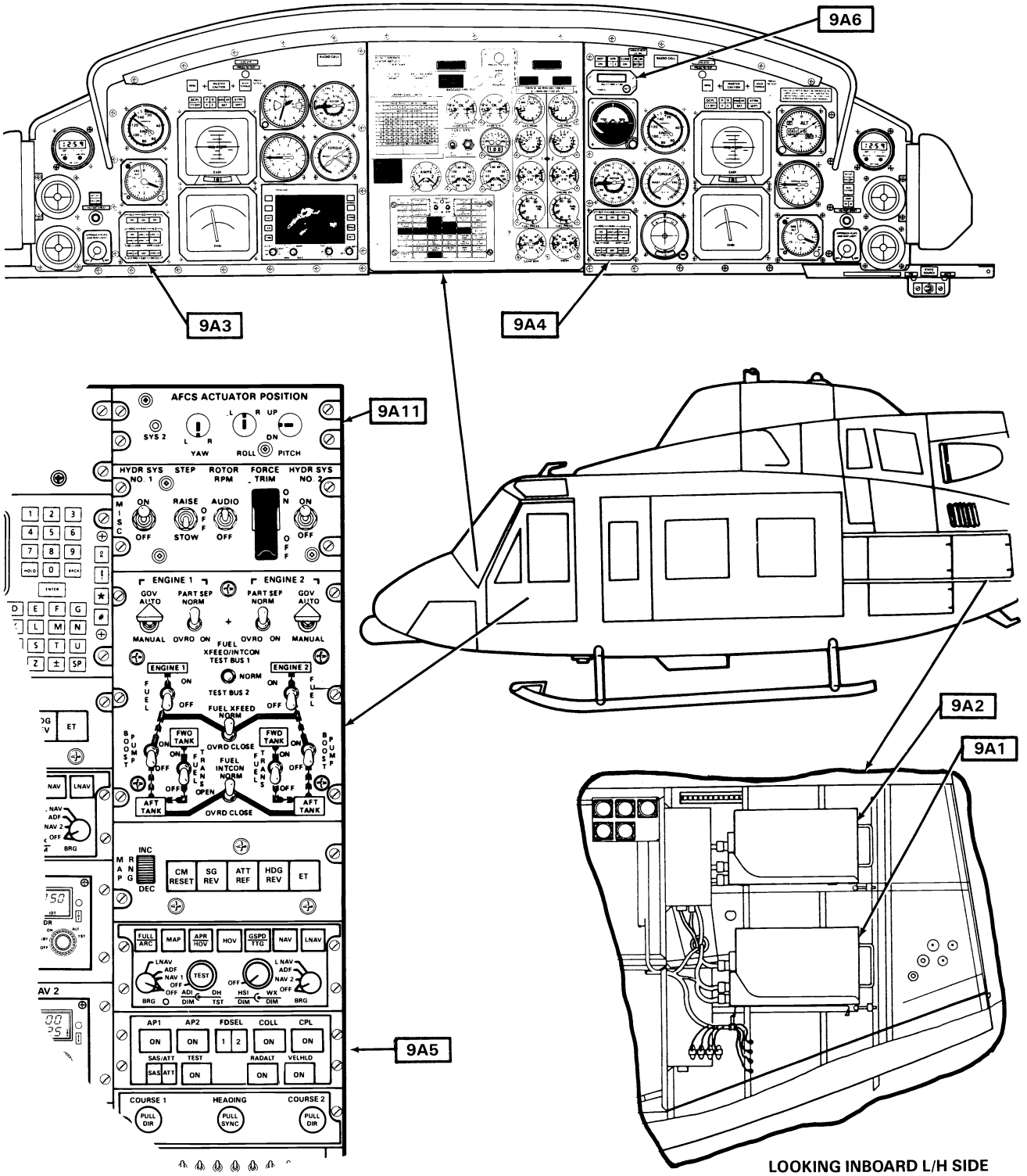



**Figure 96-6. Digital automatic flight control system — major component location**

**Table 96-2. Troubleshooting — DAFCS**

| INDICATION OF TROUBLE | PROBABLE CAUSE | CORRECTIVE ACTION |
|---|---|---|
| | **NOTE**<br><br>Error codes are displayed on the attitude preselect indicator (9A6). Refer to Honeywell, SPZ-7600 Integrated Flight Control System Bell 412HP System Maintenance Manual, Publication Number A09-1169-024 for an explaination of error codes. | |

**96-13. REMOVAL PROCEDURES.**

**WARNING**

ENSURE ELECTRICAL POWER IS OFF.

**96-14. Accelerometer assembly.**

**CAUTION**

ACCELEROMETERS ARE VERY DELICATE; HANDLE VERY CAREFULLY.

**1.** Disconnect electrical connectors.

**2.** Remove and retain mounting screws and other hardware.

**3.** Remove accelerometer assembly.

**96-15. AFCS actuator indicator panel.**

Refer to BHT-412-MM Chapter 96 for removal of pedestal mounted component.

**96-16. AFCS relay control box.**

**1.** Disconnect electrical connectors.

**2.** Remove and retain mounting screws and other hardware.

**3.** Remove relay control box.

**96-17. Air data sensor.**

**1.** Disconnect pitot and static lines.

**2.** Disconnect electrical connector.

**3.** Remove and retain mounting screws and other hardware.

**4.** Remove air data sensor.

**5.** Protect all openings with caps or tape.

**96-18. Autopilot controller.**

Refer to BHT-412-MM Chapter 96 for removal of pedestal mounted component.

**96-19. DFC computer.**

**1.** Loosen knurled nuts, at front of computer mounting rack, until holddown clamps can be slipped clear of the lug at the front of computer.

**2.** Slide computer out of its mounting rack.

**96-20. Hoist pendant control unit.**

**1.** Disconnect electrical connector.

**2.** Remove pendant control unit.

**96-21. Miscellaneous control panel.**

Refer to BHT-412-MM Chapter 96 for removal of pedestal mounted component.

**96-22. Mode selector.**

Refer to BHT-412-MM Chapter 95 for removal of panel mounted instrument.

**96-23. Yaw rate gyro.**

**CAUTION**

A GYRO MAY BE DAMAGED BY MOVING IT AFTER ELECTRICAL POWER IS REMOVED BUT BEFORE THE GYRO ROTOR STOPS. THE GYRO ROTOR WILL NOT COME TO A STOP FOR APPROXIMATELY 15 MINUTES AFTER ELECTRICAL POWER IS REMOVED. WHEN REMOVING OR INSTALLING A GYRO, DO SO GENTLY WITH NO QUICK MOTION.

**1.** Remove and retain screws and washers securing lid on yaw rate gyro assembly.

**2.** Remove lid.

**3.** Disconnect electrical connector.

**4.** Remove screws and mounting hardware.

**5.** Remove yaw rate gyro from assembly.

**96-24. INSTALLATION PROCEDURES.**

**WARNING**

ENSURE ELECTRICAL POWER IS OFF.

**96-25. Accelerometer Assembly.**

**CAUTION**

ACCELEROMETERS ARE VERY DELICATE; HANDLE VERY CAREFULLY.

**1.** Position accelerometer assembly and secure with mounting screws and other hardware.

**2.** Connect electrical connectors.

**96-26. AFCS actuator indicator panel.**

Refer to BHT-412-MM Chapter 96 for installation of pedestal mounted component.

**96-27. AFCS relay control box.**

**1.** Position control box and secure with mounting screws and other hardware.

**2.** Connect electrical connectors.

**96-28. Air data sensor.**

**1.** Position air data sensor and secure with mounting screws and other hardware.

**2.** Connect pitot and static lines.

**3.** Connect electrical connector.

**96-29. Autopilot controller.**

Refer to BHT-412-MM Chapter 96 for installation of pedestal mounted component.

**96-30. DFC computer.**

**1.** Slide computer into its mounting rack.

**2.** Lift holddown clamps, at front of mount, over the lug at front of computer.

**3.** Secure computer by tightening knurled retainer nuts.

**96-31. Hoist pendant control unit.**

**1.** Position control unit in helicopter.

**2.** Connect electrical connector.

**96-32. Miscellaneous control panel.**

Refer to BHT-412-MM Chapter 96 for installation of pedestal mounted component.

**96-33. Mode selector.**

Refer to BHT-412-MM Chapter 95 for installation of panel mounted instrument.

**96-34. Yaw rate gyro.**

**CAUTION**

WHEN REMOVING OR INSTALLING A GYRO, DO SO GENTLY WITH NO QUICK MOTION.

**1.** Position yaw rate gyro in assembly box and secure with mounting hardware.

**2.** Connect electrical connector.

**3.** Position and secure lid on assembly box.

## 96-35. INSPECTION PROCEDURES.

**1.** Inspect DAFCS for completeness, proper installation, cleanliness, and secure mounting of each component.

**2.** Inspect switches and controls on DAFCS system components for proper mechanical action.

**3.** Inspect components to ensure panel lights illuminate with electrical power on.

**4.** Inspect electrical connections for corroded or bent pins and proper mating, interconnecting cable for frayed or broken insulation, and circuit breakers for proper operation.

## 96-36. CLEANING PROCEDURES.

**1.** Remove moisture and loose dirt with a soft, clean cloth.

**2.** Remove dust and dirt from system components with a soft, clean cloth.

**WARNING**

CLEANING COMPOUND FUMES ARE FLAMMABLE AND TOXIC.

**3.** Remove grease, fungus, and ground-in dirt with a soft, clean cloth dampened with approved cleaning compound.

**4.** Remove dirt from connectors with a soft bristle brush; remove moisture with a clean, dry cloth.

## 96-37. REPAIR OR REPLACEMENT.

**1.** Tighten or replace loose or cracked control knobs or system components.

**2.** Replace defective panel light bulbs.

**3.** Replace defective or inoperative units.

**4.** Further repairs, other than removal or replacement, must be conducted at an authorized repair facility.

# ELECTRONIC FLIGHT INSTRUMENT SYSTEM

## 96-38. ELECTRONIC FLIGHT INSTRUMENT SYSTEM (EFIS).

The electronic flight instrument system (EFIS) is comprised of the following components: copilot and pilot display controllers (3440CP3 and 3440CP4), copilot and pilot electronic attitude director indicators (EADI) (3440DS1 and 3440DS2), copilot and pilot electronic horizontal situation indicators (EHSI) (3440DS3 and 3440DS4), copilot and pilot symbol generators (3440A1 and 3440A2), and remote instrument controller (3440CP1) ( figure 96-7 ).

The EFIS is a dual system that provides attitude and heading information to the pilot and copilot. The components of each system are identical and interchangeable. The EFIS displays pitch and roll attitude, heading, course orientation, weather presentations, source annunciations and doppler velocities.

The major component of the EFIS is the symbol generator. The symbol generator receives heading, attitude, radio altitude, rate of turn, doppler/velocities, short and long range navigation, and weather inputs. All inputs are processed and transmitted to the electronics displays as a function of the selections made on the display controller and the remote instrument controller. If one symbol generator fails, backup is provided by the other symbol generator. The symbol generators communicate over the avionics standard communications bus (ASCB) to perform comparison monitoring.

## 96-39. OPERATIONAL CHECK — SAR EFIS.

**1.** Visually inspect equipment and correct any faulty mechanical mountings or electrical connections. Connect external power, adjusted to 27.5 Vdc, to helicopter power receptacle.

**2.** Energize power buses per paragraph 97-4.

**3.** Apply external hydraulic pressure, refer to BHT-412-MM , Chapter 29 .

**4.** Close EFIS No. 1 circuit breakers ADI (3440CB1), HSI (3440CB3), SG/DL (3440CB5), EFIS 26V NO. 1 (3440CB7), BLWR (3440CB9), and EFIS No. 2 circuit breakers ADI (3440CB2), HSI (3440CB4), SG/DL (3440CB6), EFIS 26V NO. 2 (3440CB8), and BLWR (3440CB10).

**5.** Close circuit breakers AFCS 26V NO. 1 (9CB4), RATE GYRO NO. 1 (9CB31), AFCS NO. 1 PWR (9CB31), AFCS NO. 1 ADD/CAUT (9CB32), and FORCE TRIM (9CB3).

**6.** Close circuit breakers AFCS 26V NO. 2 (9CB7), RATE GYRO NO. 2 (9CB30), AFCS NO. 2 PWR (9CB2), AFCS NO. 2 CAUT (9CB33), and ENC ALT (3410CB1).

**7.** Set EFIS MASTER switch to ON.

**8.** Perform operational checkout according to Honeywell, SPZ-7600 Integrated Flight Control System Bell 412HP System Maintenance Manual, Publication Number A09-1169-024.

## 96-40. TROUBLESHOOTING.

It is assumed that all appropriate circuit breakers are closed, interconnecting cabling is in normal operational condition, and power is applied to the system. (See appropriate wiring diagram.) Refer to Honeywell, SPZ-7600 Integrated Flight Control System Bell 412HP System Maintenance Manual, Publication Number A09-1169-024 for troubleshooting.

## 96-41. REMOVAL PROCEDURES.

**WARNING**

ENSURE ELECTRICAL POWER IS OFF.

**96-42. Auxiliary switch panel.**

Refer to BHT-412-MM Chapter 96 for removal of pedestal mounted component.

**96-43. Display controller.**

Refer to BHT-412-MM Chapter 96 for removal of pedestal mounted component.

**96-44. Instrument controller.**

Refer to BHT-412-MM Chapter 96 for removal of pedestal mounted component.

**96-45. Symbol generator.**

**1.** Loosen mounting tray knob, at front of symbol generator.

**2.** Slide symbol generator out of its mounting rack.

**96-46. INSTALLATION PROCEDURES.**

**WARNING**

ENSURE ELECTRICAL POWER IS OFF.

**96-47. Auxiliary switch panel.**

Refer to BHT-412-MM Chapter 96 for installation of pedestal mounted component.

**96-48. Display controller.**

Refer to BHT-412-MM Chapter 96 for installation of pedestal mounted component.

**96-49. Instrument controller.**

Refer to BHT-412-MM Chapter 96 for installation of pedestal mounted component.

**96-50. Symbol generator.**

**CAUTION**

DO NOT FORCE FIT. IF MATING IS DIFFICULT, REMOVE THE UNIT AND CHECK FOR CONNECTOR PINS THAT MAY BE BENT OR OUT OF ALIGNMENT. ALSO CHECK THE ALIGNMENT OF THE RECEPTACLE IN THE MOUNTING TRAY.

**1.** Slide symbol generator into its mounting rack.

**2.** Tighten mounting tray knob, at front of symbol generator.

**96-51. INSPECTION PROCEDURES.**

**1.** Inspect electronic flight instrument system for completeness, proper installation, cleanliness, and secure mounting of each component.

**2.** Inspect switches and controls on electronic flight instrument system components for proper mechanical action.

**3.** Inspect components to ensure panel lights illuminate with electrical power on.

**4.** Inspect electrical connections for corroded or bent pins and proper mating, interconnecting cable for frayed or broken insulation, and circuit breakers for proper operation.

**96-52. CLEANING PROCEDURES.**

**1.** Remove moisture and loose dirt with a soft, clean cloth.

**2.** Remove dust and dirt from system components with a soft, clean cloth.

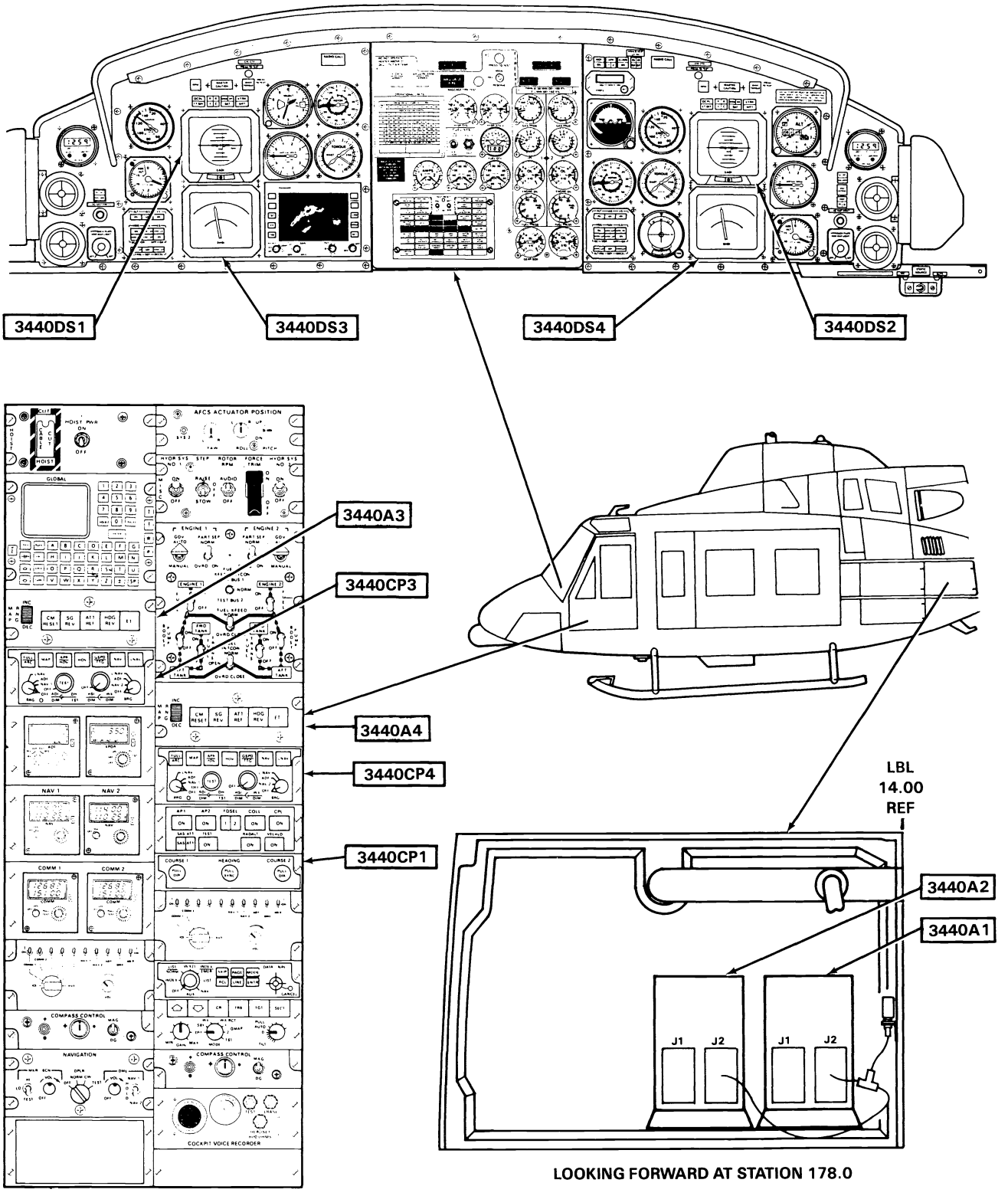



**Figure 96-7. Electronic flight instrument system — component location**

**WARNING**

CLEANING COMPOUND FUMES ARE FLAMMABLE AND TOXIC.

**3.** Remove grease, fungus, and ground-in dirt with a soft, clean cloth dampened with approved cleaning compound.

**4.** Remove dirt from connectors with a soft bristle brush; remove moisture with a clean, dry cloth.

**96-53. REPAIR OR REPLACEMENT.**

**1.** Tighten or replace loose or cracked control knobs or system components.

**2.** Replace defective panel light bulbs.

**3.** Replace defective or inoperative units.

**4.** Further repairs, other than removal or replacement, must be conducted at an authorized repair facility.